Software Engineering Document

Customer: Home Depot

Student Group 5

Company Logo: BuildSmart

Date: March 6th, 2026

Team Members: Prajit Alexander, Nathan Tennyson, Alen Jo, Estrella Avila

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Introduction of Document

This document expands upon Project Assignment 1 and provides the structured technical specifications for Build Smart: A Home Depot AI Project-to-Purchase Planning Service.

This document formally defines the system requirements, architecture considerations, and testing strategies necessary to guide the development of our project. This document focuses on key software engineering applications like requirements analysis, system modeling, architecture design, and test planning. Through stakeholder perspectives, functional and non-functional requirements, and system modeling artifacts, this document outlines how the system is expected to operate and interact with users and external services.

Requirements / System Modeling

Stakeholder and Team Member	Stakeholder Question	Functional Requirement	Non-Functional Requirement
Director of Customer Relations (Estrella)	1. Should customers be able to save their AI-generated home improvement plans so they can revisit them later? 2. Should customers have the ability to share their generated project plans with family members, friends, or contractors for collaboration? 3. How can the system help customers easily access their project plans when they visit a store and request assistance? 4. Should the system generate a clear	1. The system shall allow users to save and revisit previously generated project plans through a user account dashboard. 2. The system shall allow users to share generated project plans with others through a shareable link or exportable document. 3. The system shall allow in-store associated to search and retrieve AI-generated project plans to assist customers during consultations.	1. The system shall maintain 99.5% uptime availability to ensure users can access project planning services reliably throughout the year. 5. The user interface shall allow users to generate a complete project plan within three steps or fewer, ensuring that customers with limited technical experience can easily explore recommended or trending projects.

	shopping list that customers can follow while navigating a physical Home Depot store? 5. Should the system recommend trending or popular home improvement projects to help inspire customers who are unsure where to start?	4. The system shall generate a printable or mobile-friendly shopping list that customers can use while navigating a physical Home Depot store. 5. The system shall recommend home improvement projects based on seasonal demand and customer behavior analytics.	
In-Store Sales Associate (Estrella)	1. Should the system display any promotions or discounts related to recommended materials, so associates can inform customers about savings opportunities? 2. How should the system ensure recommended products are available at the customer's selected store location? 3. What should happen if a recommended product is unavailable at the store? Should alternative products be suggested automatically?	1. The system shall display promotional offers discounts for recommended materials within the generated project plans. 2. The system shall prioritize recommended products that are currently in stock at the user's selected Home Depot store location. 3. The system shall prioritize recommended when a required item is out of stock. 4. The system shall collect anonymized usage data on generated	2. The system shall automatically recover from temporary service interruptions and restore user sessions without loss of project plans or inventory data, ensuring associates can continue assisting customers without disruption. 3. The system architecture shall use modular components, so product availability and alternative recommendations can be updated without affecting the entire system.

	4. Should the system collect anonymized data about generated project plans to help improve future customer recommendations? 5. Should the system generate reports showing the most common types of projects customers request so store teams can prepare inventory and materials?	project plans to support marketing and product strategy analysis. 5. The system shall generate reports summarizing the most frequently requested project types and materials.	
Marketing Manager (Prajit)	1. How can we market the simplicity of “describe your project” in plain language to attract the non-technical DIY users? 2. What visual assets from user-uploaded photos can we ethically leverage to showcase real-world project transformations in our marketing materials? 3. How can we balance the promotion of DIY material sales compared to the “Find a Pro” referral service to maximize overall revenue without	1. The system shall use Natural Language Processing to extract project scope, materials, and intent from free-text user descriptions. 2. The system shall allow users to upload photos of their current space or hand-drawn sketches to automatically identify layout constraints and existing structures. 3. The system shall provide a "Find a Pro" button for complex project phases that links the current Bill of Materials and project plan to	1. The backend architecture shall support horizontal scaling to handle up to 50,000 concurrent users during peak sales events without a loss in performance. 2. The system shall encrypt all user-uploaded photos and personal data at rest using Advanced Encryption Standard-256 standards to comply with the General Data Protection Regulation. 3. The system shall ensure that any project plan completed using the web interface is accessible and

	confusing the customer. 4. How can we use the system's seasonal material alerts to drive localized, climate-specific marketing campaigns during extreme weather events such as winterization? 5. In what ways can the 3D AR preview feature be positioned in our social media advertising to reduce "buyer's remorse" and increase conversion rates?	Home Depot's authorized service providers. 4. The system shall notify users if specific project materials are seasonal (e.g., exterior paint in winter) and suggest climate-appropriate alternatives or timing. 5. The system shall generate a 3D AR model of the planned project that users can overlay onto their physical space using a mobile device camera.	consistent on a mobile application within 2 seconds of a manual save.
Digital Analytics Specialist (Prajit)	1. What metrics should we track to ensure users aren't abandoning their carts due to the automated calculation of localized taxes or environmental fees? 2. How will we measure the conversion rate for "Pro-Loader" assistance requests versus traditional bulk delivery slots to optimize store staffing data? 3. What specific data points are needed	1. The system shall calculate final project costs including local sales tax and environmental disposal fees based on the user's specific zip code. 2. The system shall provide available delivery slots for bulk materials and allow users to schedule "Pro-Loader" assistance for in-store pickups.	2. The system shall implement Multi Factor Authentication for all Admin and Manager accounts and utilize Open Authorization 2.0 for Customer social logins.

	to track the retention rate and long-term "stickiness" of users who set up recurring orders for project consumables? 4. How should the system report the impact of automatic volume pricing milestones on the Total Lifetime Value (LTV) of our Pro Xtra members? 5. What data do we need to collect on split-payment usage to determine if offering financing plans directly in the tool leads to higher average order values?	3. The system shall allow users to set up recurring orders for project-related consumables after project completion. 4. The system shall automatically apply member discounts and track project spending toward Pro Xtra rewards or volume pricing milestones. 5. The system shall allow users to split project payments across Home Depot gift cards, credit cards, and financing plans.	
Product Category Manager (Alen)	1. When a customer comes in for a particular project, how confident are you that they're actually leaving with everything they need and how do you know if they're not confident? 2. If you wanted to make a quick swap on what's recommended or promoted in a specific category, what does that process look	1. The system shall allow users to input a home improvement project description in natural language and receive a structured project plan in response. 2. The system shall map every step of a project plan to specific, purchasable Home Depot SKUs with real-time pricing.	1. The attach-rate recommendation engine shall return at least three complementary SKUs per project step with a strong relevance score that's statistically sound from Home Depot's catalog offerings.

	like today and how long does it usually take? 3. How do customers typically communicate what they want to build and does it match the expectations to Home Depot's catalog offerings?	3. The system shall generate a complete, itemized cart that users can review and transfer directly to HomeDepot.com for checkout. 5. The system shall display estimated total project cost, broken down by materials, tools, and optional add-ons. 6. The system shall surface attach-rate recommendations (e.g., "customers who buy drywall also need...") to improve basket completeness. 7. The system shall allow users to filter, or swap recommended products by brand, price range, or availability (in-store vs. online).	
Inventory / Supply Chain Manager (Alen)	1. Right now, how are you getting input on what DIYers are planning to buy before they actually show up is it early enough to be good for their project planning? 2. When a big seasonal push hits and demands spike on certain inventory, what is the biggest breakdown point	2. The system shall map every step of a project plan to specific, purchasable Home Depot SKUs with real-time pricing. 4. The system shall provide quantity recommendations for all materials based on user-provided dimensions or project scope.	2. The system shall sync product pricing, availability, and SKU data from the Home Depot catalog API at intervals not exceeding 15 minutes. 3. The demand signal pipeline shall properly batch and export recommended SKU, quantities, and relevant metrics to Home

	between what customers expect to find versus what they actually find? 3. How are you seeing returns tied to customers buying the wrong items or wrong quantities and do you know where in their project planning stage they went wrong to have this happen to them?	8. The system shall check and display real-time product availability by store location or zip code. 9. The system shall provide step-by-step project guidance alongside each recommended product to improve DIY confidence and reduce misuse-related returns. 10. The system shall export demand signal data (recommended SKUs, quantities, frequency) to Home Depot's inventory/demand planning systems to support supply chain forecasting.	Depot's BI system in a 24-hour period.
IT Systems Architect (Nathan)	1. What key system metrics should be visible on a health dashboard for the operations team? 2. Should the health dashboard display alerts when something goes wrong with the system? 3. What user and system events should the application log? 4. Should administrators be	1. The system shall provide a health dashboard that displays real-time metrics including server uptime, API response times, and active user sessions. 2. The system shall display alerts on the health dashboard when a monitored metric exceeds a defined threshold.	1. The system shall generate system health alerts within 30 seconds of detecting a critical service failure to support timely incident response. 2. The system shall retain system logs and audit records for a minimum of 12 months to support troubleshooting, compliance, and security investigations

	able to search and filter through logs in the log viewer? 5. Does the app need to integrate with any exisiting Home Depot monitoring tools?	3. The system shall log all significant user and system events, including login attempts, project updates, and application errors, with a timestamp and user ID recorded for each entry. 4. The system shall provide a log viewer that allows administrators to search and filter log entries by date, user ID, and event type. 5. The system shall support data export from the health dashboard and log viewer to external monitoring tools used by the Home Depot IT.	
Data Privacy Officer (Nathan)	1. What categories of customer data will this app collect, and what consent mechanisms must be in place before that data is stored or processed? 2. Are there specific data retention policies we must enforce. For example,, how long can a user's project history, saved items or	1. The system shall require clear user consent before collecting, storing, or processing any personally identifiable information (PII), and this consent shall be presented through an in-app prompt during	1. The system shall encrypt all sensitive user data, including uploaded photos and location data, both in transit and at rest using industry-standard 2. The system shall comply with applicable data privacy regulations such as GDPR and CCPA when collecting, storing,

	personal information be stored? 3. What user rights must the app support, such as ability to request data deletion, data export, or opt out of personalized recommendations? 4. Are there regulatory frameworks that apply to our user base, and what specific compliance controls must be built into the application? 5. How must the app handle a data breach scenario, are the notification workflows or audit loggin requirements that ened to be built into the system?	account registration. 2. The system shall enforce configurable data retention policy that automatically flags or deletes user project data, saved materials lists, and account information after a defined period of inactivity. 3. The system shall provide a self-service Privacy Center where users can view, export, or permanently delete their personal data, with deletion requests completed within 30 days to comply with the California Consumer Privacy Act. 1. The system shall provide an opt-out option that allows users to disable personalized product recommendations and behavioral data tracking at any time from	and processing user data.

		their account settings. 2. The system shall generate and store secure audit logs of all user data access, changes, and deletions, and retain these logs for at least 12 months to support investigations and regulatory reporting.	

User Stories

ID	User Story	Related FRQ/s	Team Member
US-1	As a customer, I want to be asked for consent before my data is used for analytics or marketing, so that I can control how my personal information is used by the system.	FR-31	Nathan Tennyson
US-2	As a customer, I want to review and modify my privacy and consent settings, so that I can manage how my project data and activity information are used by the platform.	FR-33, FR-34	Nathan Tennyson
US-3	As a system administrator, I want to view system health metrics and receive alerts when critical services fail, so that I can quickly identify and resolve issues that could affect users.	FR-36, FR-37	Nathan Tennyson
US-4	As a support engineer, I want to access system logs and service events, so that I can troubleshoot errors and investigate system incidents efficiently.	FR-38, FR-39, FR-40	Nathan Tennyson
US-5	As a busy homeowner, I want to upload a photo of my room and describe my project intent so that the system can automatically identify layout constraints and generate an accurate material list without manual data entry.	FR-11, FR-12	Prajit Alexander
US-6	As a DIY customer, I want to calculate my total project costs including local taxes and split the final payment across multiple gift cards and credit cards so that I can manage my project finances accurately before scheduling my pickup.	FR-13, FR-14, FR-20	Prajit Alexander
US-7	As a novice builder, I want to visualize my project in my actual space using augmented reality and then click the "Find a Pro" button so that I can verify that the design looks correct before hiring a contractor to help me with the installation phases.	FR-16, FR-19	Prajit Alexander
US-8	As a Pro Xtra member, I want to receive seasonal material alerts for my completed projects and set up recurring subscriptions for filters and cleaners so that I can maintain my investment while automatically earning loyalty rewards on every maintenance purchase.	FR-15, FR-17, FR-18	Prajit Alexander
US-9	As a homeowner planning a project over multiple days, I want to save my generated project plans and	FR-21, FR-22	Estrella Avila

	share it with family, friends, or contractors so that I can revisit them later before purchasing the materials.		
US-10	As a Home Depot associate, I want to retrieve the customer's project plans and generate a shopping list, so that I can quickly help them locate the required materials in the store.	FR-23, FR-24	Estrella Avila
US-11	As a customer shopping for project materials, I want the system to prioritize items that are in stock at my selected store and suggest alternatives if items are unavailable, so that I can complete my project without delays.	FR-27, FR-28	Estrella Avila
US-12	As a homeowner looking for project inspiration, I want to see trending home improvement projects along with current promotions, so that I can discover new ideas while saving money.	FR-25, FR-26	Estrella Avila
US-13	As a business analyst, I want the system to collect anonymized data on generated project plans and produce reports on popular projects and materials, so that the company can improve inventory planning and marketing strategies.	FR-29, FR-30	Estrella Avila
US-14	As a homeowner without prior construction experience, I should be able to describe my project in plain, everyday language (like "I need to tile a 12x10 bathroom floor) and should receive a structured, step-by-step plan so that I can understand exactly what the job involves without needing to do research myself.	FR-1, FR-4	Alen Jo
US-15	As a DIY shopper, I want every material in my project plan mapped to a specific, purchasable Home Depot product with live pricing and can transfer that product into HomeDepot.com's shopping cart, so that I can go from an idea to checkout without manually looking up items.	FR-2, FR-3	Alen Jo
US-16	As a budget-friendly homeowner, I want to see my total estimated quote broken down by materials, tools, optional upgrades, and commonly forgotten items, so that I can plan my budget as accurately as I can make it and avoid many trips to Home Depot.	FR-5, FR-6	Alen Jo
US-17	As a shopper with specific preferences, I want to filter products by brand and availability while being able to see real-time stock at my nearest Home Depot or zip	FR-7, FR-8	Alen Jo

	code, so that I can substitute items that fit my preferences and pick up what I need today.		
US-18	As a first-time DIYer, I want step-by-step instructions displayed along recommended items so that I can complete my project correctly and avoid returns	FR 9	Alen Jo
US-19	As a Home Depot inventory manager, I want aggregated SKUs and quantity data from all user sessions exported to a BI (Business Intelligence) system so that I can report which items are in-demand and need to be stocked	FR 10	Alen Jo

Functional Requirements

(Alen) FR-1: The system shall allow users to input a home improvement project description in natural language and receive a structured project plan in response.

(Alen) FR-2: The system shall map every step of a project plan to specific, purchasable Home Depot SKUs with real-time pricing.

(Alen) FR-3: The system shall generate a complete, itemized cart that users can review and transfer directly to HomeDepot.com for checkout.

(Alen) FR-4: The system shall provide quantity recommendations for all materials based on user-provided dimensions or project scope.

(Alen) FR-5: The system shall display estimated total project cost, broken down by materials, tools, and optional add-ons.

(Alen) FR-6: The system shall surface attach-rate recommendations (e.g., "customers who buy drywall also need...") to improve basket completeness.

(Alen) FR-7: The system shall allow users to filter, or swap recommended products by brand, price range, or availability (in-store vs. online).

(Alen) FR-8: The system shall check and display real-time product availability by store location or zip code.

(Alen) FR-9: The system shall provide step-by-step project guidance alongside each recommended product to improve DIY confidence and reduce misuse-related returns.

(Alen) FR-10: The system shall export demand signal data (recommended SKUs, quantities, frequency) to Home Depot's inventory/demand planning systems to support supply chain forecasting.

(Prajit) FR-11: System shall use Natural Language Processing to extract project scope, materials, and intent from free-text user descriptions.

(Prajit) FR-12: System shall allow users to upload photos of their current space or hand-drawn sketches to automatically identify layout constraints and existing structures.

(Prajit) FR-13: The system shall calculate final project costs including local sales tax and environmental disposal fees based on the user's specific zip code.

(Prajit) FR-14: The system shall provide available delivery slots for bulk materials and allow users to schedule "Pro-Loader" assistance for in-store pickups.

(Prajit) FR-15: The system shall allow users to set up recurring orders for project-related consumables after project completion.

(Prajit) FR-16: The system shall provide a "Find a Pro" button for complex project phases that links the current Bill of Materials and project plan to Home Depot's authorized service providers.

(Prajit) FR-17: The system shall notify users if specific project materials are seasonal (e.g., exterior paint in winter) and suggest climate-appropriate alternatives or timing.

(Prajit) FR-18: The system shall automatically apply member discounts and track project spending toward Pro Xtra rewards or volume pricing milestones.

(Prajit) FR-19: The system shall generate a 3D AR model of the planned project that users can overlay onto their physical space using a mobile device camera.

(Prajit) FR-20: The system shall allow users to split project payments across Home Depot gift cards, credit cards, and financing plans.

(Estrella) FR-21: The system shall allow users to save and revisit previously generated project plans through a user account dashboard.

(Estrella) FR-22: The system shall allow users to share generated project plans with others through a shareable link or exportable document.

(Estrella) FR-23: The system shall allow in-store associates to search and retrieve AI-generated project plans to assist customers during consultations.

(Estrella) FR-24: The system shall generate a printable or mobile-friendly shopping list that customers can use while navigating a physical Home Depot store.

(Estrella) FR-25: The system shall recommend popular or trending home improvement projects based on seasonal demand and customer behavior analytics.

(Estrella) FR-26: The system shall display promotional offers or discounts for recommended materials within the generated project plans.

(Estrella) FR-27: The system shall prioritize recommended products that are currently in stock at the user's selected Home Depot store location.

(Estrella) FR-28: The system shall provide alternative product recommendations when a required item is out of stock.

(Estrella) FR-29: The system shall collect anonymized usage data on generated project plans to support marketing and product strategy analysis.

(Estrella) FR-30: The system shall generate reports summarizing the most frequently requested project types and materials.

(Nathan) FR-31: The system shall require clear user consent before collecting, storing, or processing any personally identifiable information (PII), and this consent shall be presented through an in-app prompt during account registration.

(Nathan) FR-32: The system shall enforce configurable data retention policy that automatically flags or deletes user project data, saved materials lists, and account information after a defined period of inactivity.

(Nathan) FR-33: The system shall provide a self-service Privacy Center where users can view, export, or permanently delete their personal data, with deletion requests completed within 30 days to comply with the California Consumer Privacy Act.

(Nathan) FR-34: The system shall provide an opt-out option that allows users to disable personalized product recommendations and behavioral data tracking at any time from their account settings.

(Nathan) FR-35: The system shall generate and store secure audit logs of all user data access, changes, and deletions, and retain these logs for at least 12 months to support investigations and regulatory reporting.

(Nathan) FR-36: The system shall provide a health dashboard that displays real-time metrics including server uptime, API response times, and active user sessions.

(Nathan) FR-37: The system shall display alerts on the health dashboard when a monitored metric exceeds a defined threshold.

(Nathan) FR-38: The system shall log all significant user and system events, including login attempts, project updates, and application errors, with a timestamp and user ID recorded for each entry.

(Nathan) FR-39: The system shall provide a log viewer that allows administrators to search and filter log entries by date, user ID, and event type.

(Nathan) FR-40: The system shall support data export from the health dashboard and log viewer to external monitoring tools used by the Home Depot IT.

Non-Functional Requirements

(Alen) NFR-1: The attach-rate recommendation engine shall return at least three complementary SKUs per project step with a strong relevance score that's statistically sound from Home Depot's catalog offerings.

(Alen) NFR-2: The system shall sync product pricing, availability, and SKU data from the Home Depot catalog API at intervals not exceeding 15 minutes.

(Alen) NFR-3: The demand signal pipeline shall properly batch and export recommended SKU, quantities, and relevant metrics to Home Depot's BI system in a 24-hour period.

(Prajit) NFR-4: The system shall encrypt all user-uploaded photos and personal data at rest using Advanced Encryption Standard-256 standards to comply with the General Data Protection Regulation.

(Prajit) NFR-5: The system shall implement Multi-Factor Authentication for all Admin and Manager accounts and utilize Open Authorization 2.0 for customer social logins.

(Prajit) NFR-6: The backend architecture shall support horizontal scaling to handle up to 50,000 concurrent users during peak sales events without a loss in performance.

(Estrella) NFR-7: The system shall maintain 99.5% uptime availability to ensure users can access project planning services reliably throughout the year.

(Estrella) NFR-8: The user interface shall allow users to generate a complete project plan within three steps or fewer, ensuring that customers with limited technical experience can easily explore recommended or trending projects.

(Estrella) NFR-9: The system shall automatically recover from temporary service interruptions and restore user sessions without loss of project plans or inventory data, ensuring associates can continue assisting customers without disruption.

(Estrella) NFR-10: The system architecture shall use modular components, so product availability and alternative recommendations can be updated without affecting the entire system.

(Nathan) NFR-11: The system shall generate system health alerts within 30 seconds of detecting a critical service failure to support timely incident response.

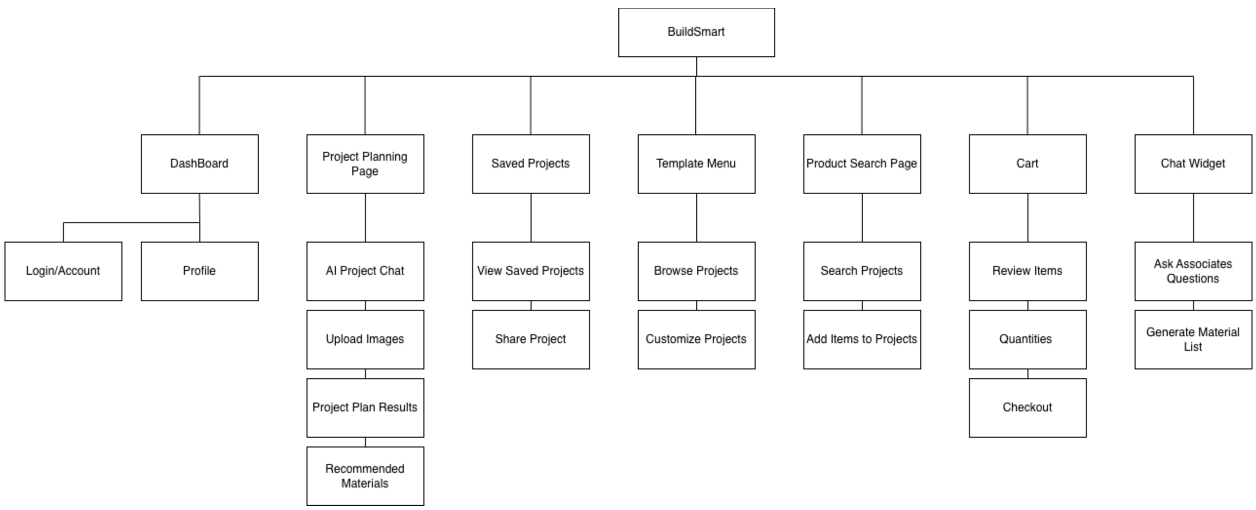
(Nathan) NFR-12: The system shall retain system logs and audit records for a minimum of 12 months to support troubleshooting, compliance, and security investigations

(Nathan) NFR-13: The system shall encrypt all sensitive user data, including uploaded photos and location data, both in transit and at rest using industry-standard encryption protocols.

(Nathan) NFR-14: The system shall comply with applicable data privacy regulations such as GDPR and CCPA when collecting, storing, and processing user data.

(Prajit) NFR-15: The system shall ensure that any project plan completed using the web interface is accessible and consistent on a mobile application within 2 seconds of a manual save.

Site Map

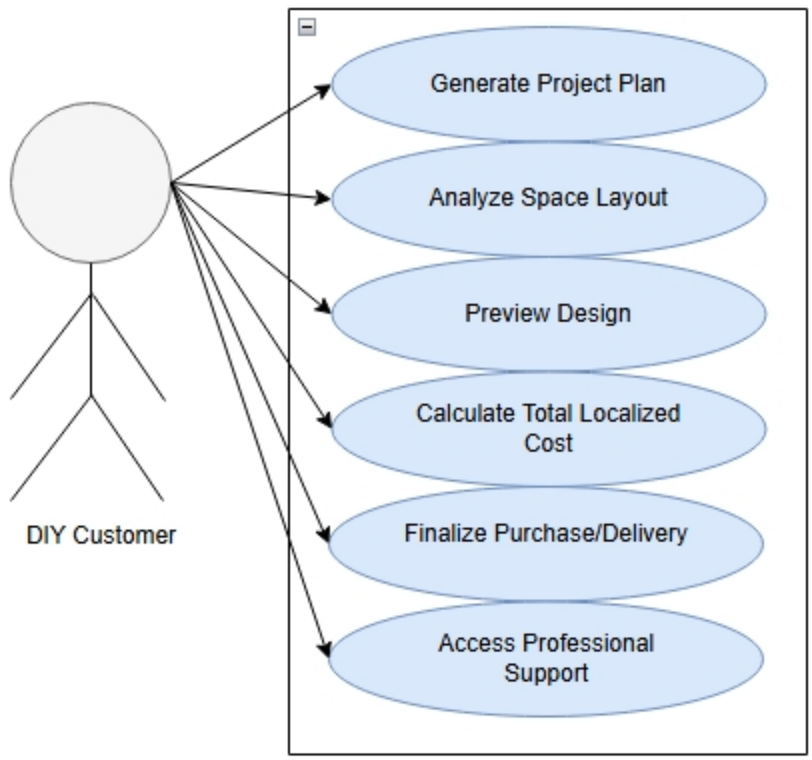


The site map shows the navigation of the BuildSmart platform with major components of the system interface.

Use Case Diagrams

(Prajit Alexander, FR-11 through FR-20)

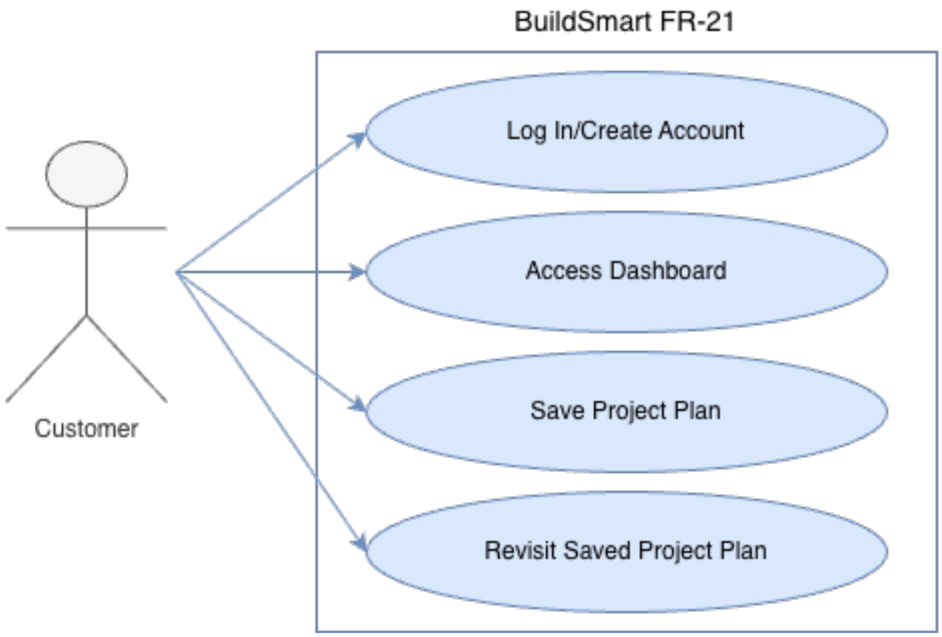
The use case diagram serves as a visual roadmap of how different users interact with the systems core features to turn a project idea into a complete purchase.



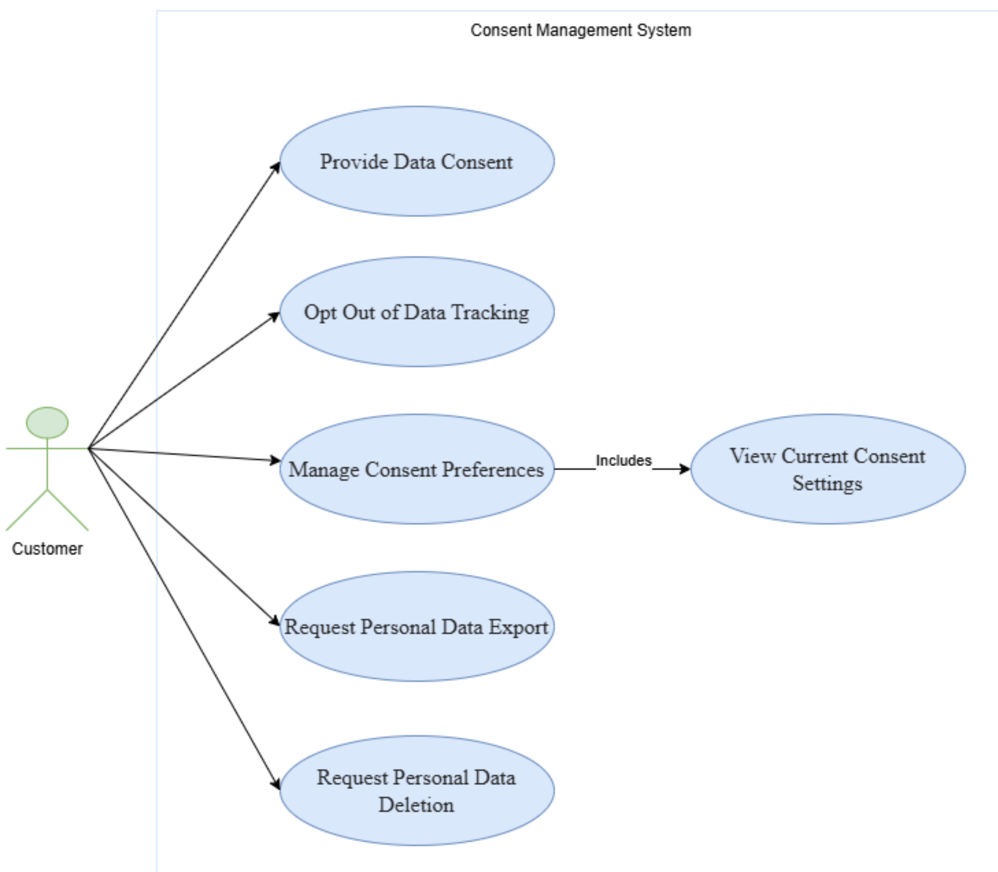
DIY Customers are the primary actors who will provide the input for the system. They can describe their project and upload photos of their space. The system then generates a 3-D Augmented Reality Preview to allow the user to visualize the result before buying the materials. The total cost of the project is then calculated while taking the user's local area or residency and Home Depot membership into account to apply taxes or discounts. The delivery is then finalized as the user inputs their payment method and the transaction goes through. The user also has an option to be referred to professional help after the payment has been made.

(Estrella Avila FR-21)

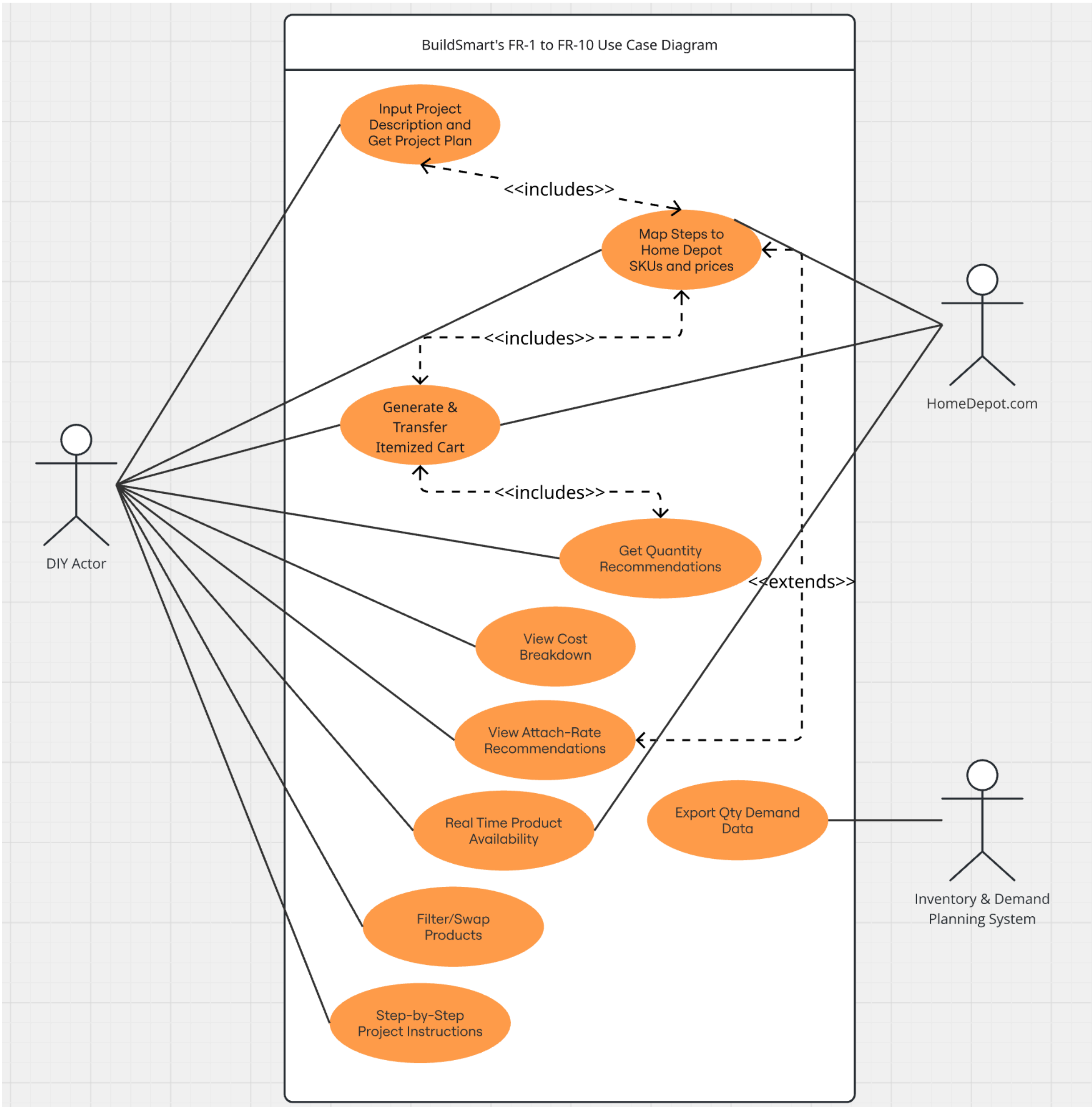
This use case diagram illustrates how a **Customer** interacts with the system to save an AI-generated project plan. The customer must log in or create an account, after which they can access their dashboard to save and later revisit previous project plans.



Use Case Diagram 3 – Consent Management (Nathan Tennyson, FR-31 through FR-40)

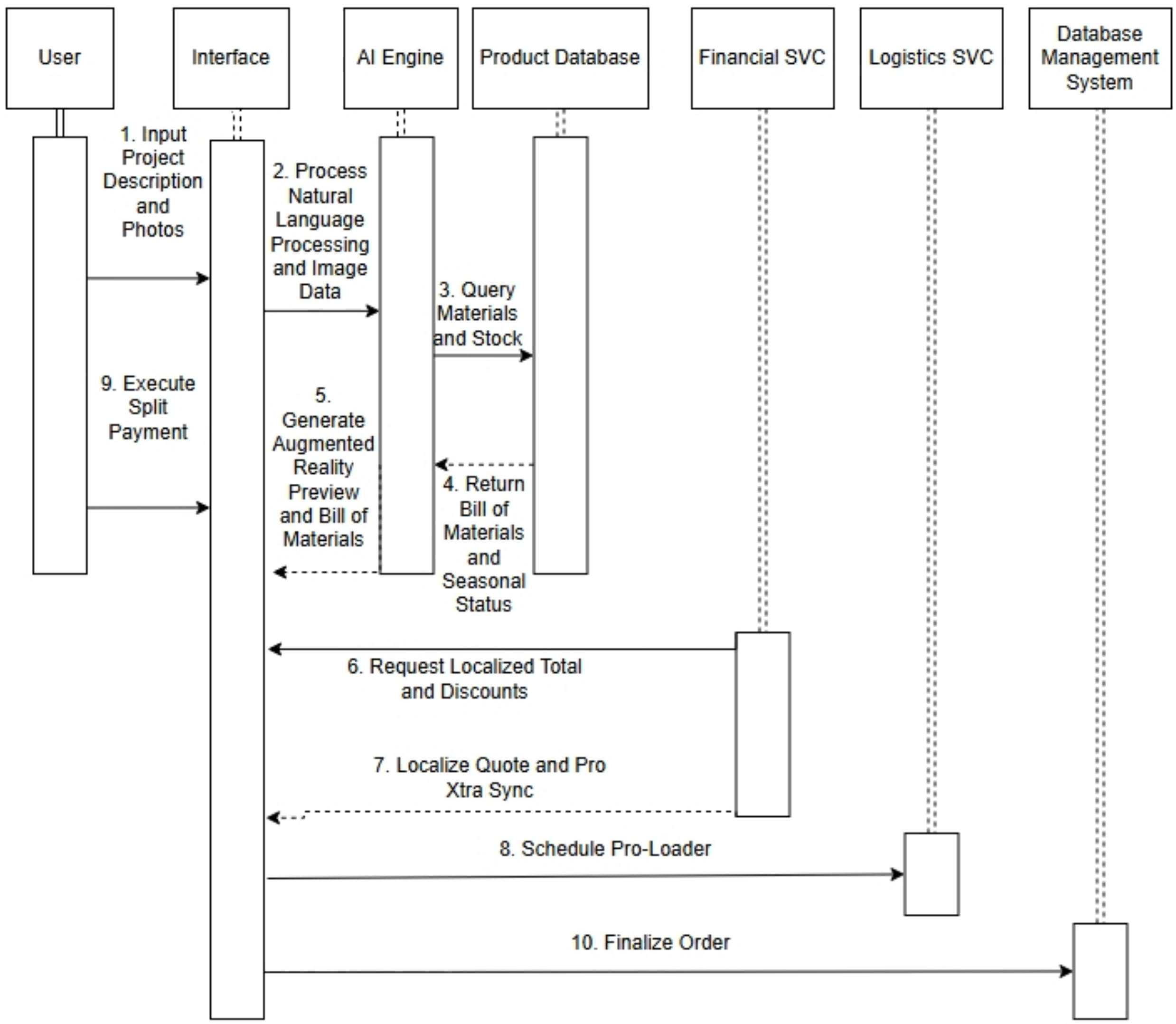


(Alen Jo, FR #1 to FR #10)

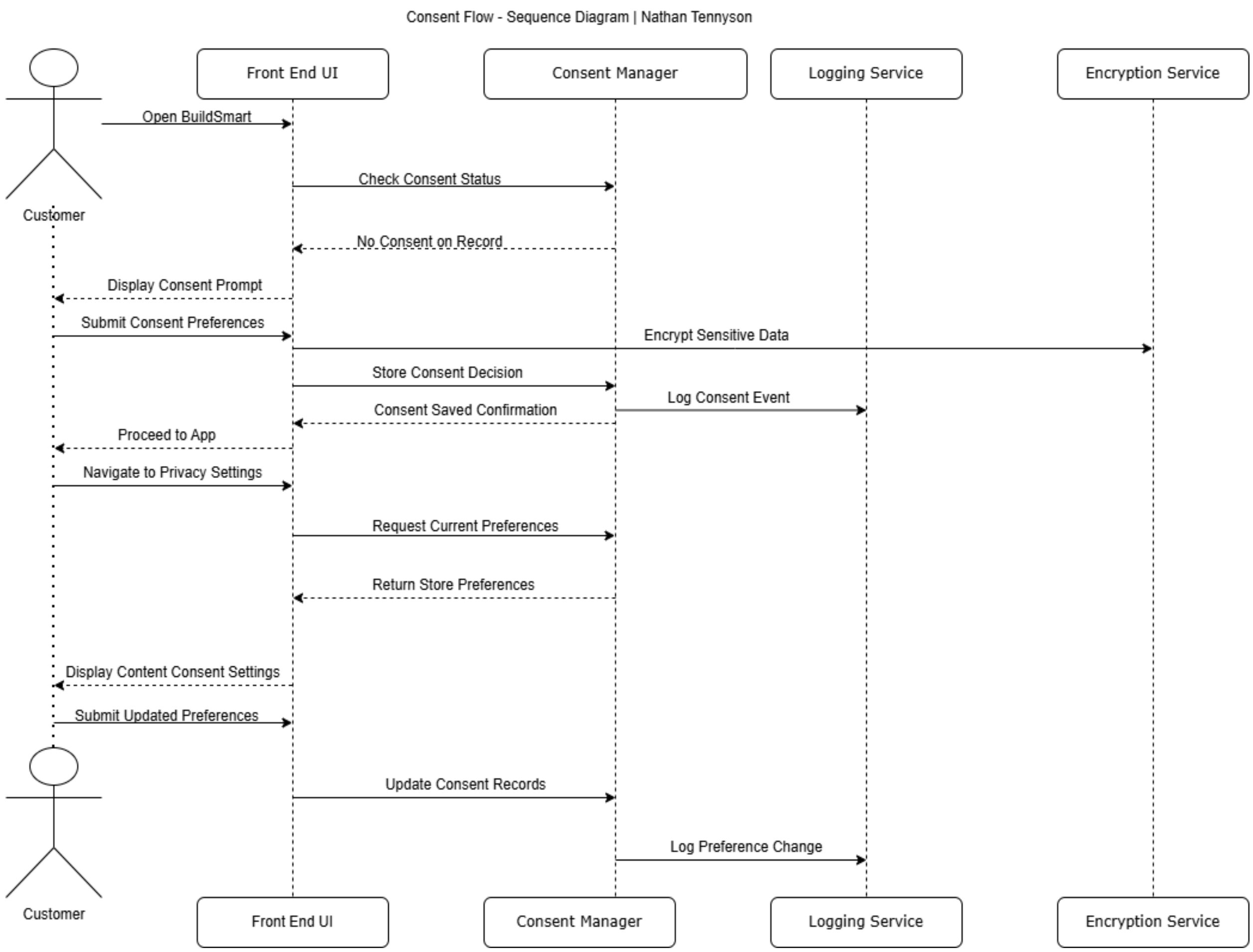


Sequence Diagrams

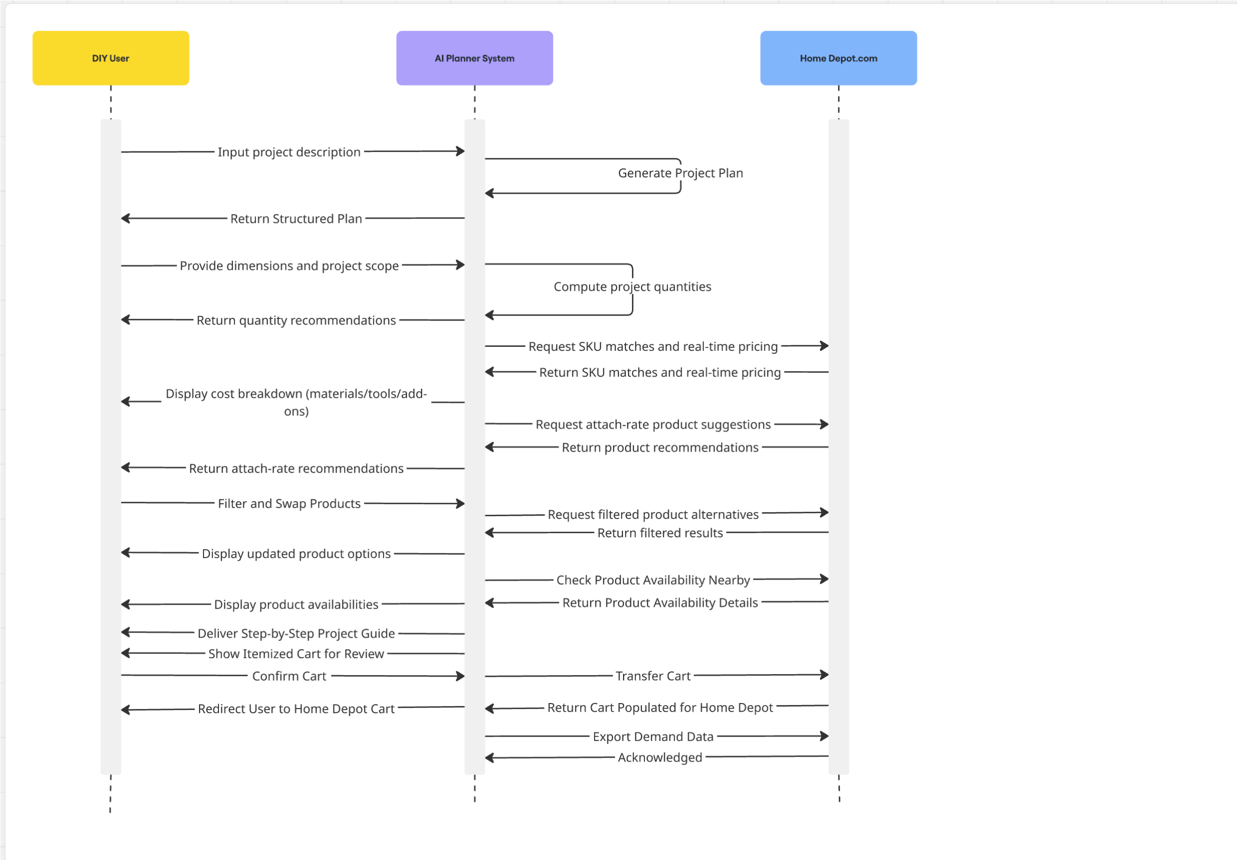
(Prajit Alexander, FR-11 through FR-20)


User
Interface
AI Engine
Product Database
Financial SVC
Logistics SVC
Database Management System
1. Input Project Description and Photos
2. Process Natural Language Processing and Image Data
3. Query Materials and Stock
4. Return Bill of Materials and Seasonal Status
5. Generate Augmented Reality Preview and Bill of Materials
6. Request Localized Total and Discounts
7. Localize Quote and Pro Xtra Sync
8. Schedule Pro-Loader
9. Execute Split Payment
10. Finalize Order

Sequence Diagram 2 - Consent Flow (Nathan Tennyson, FR-31 through FR-40)

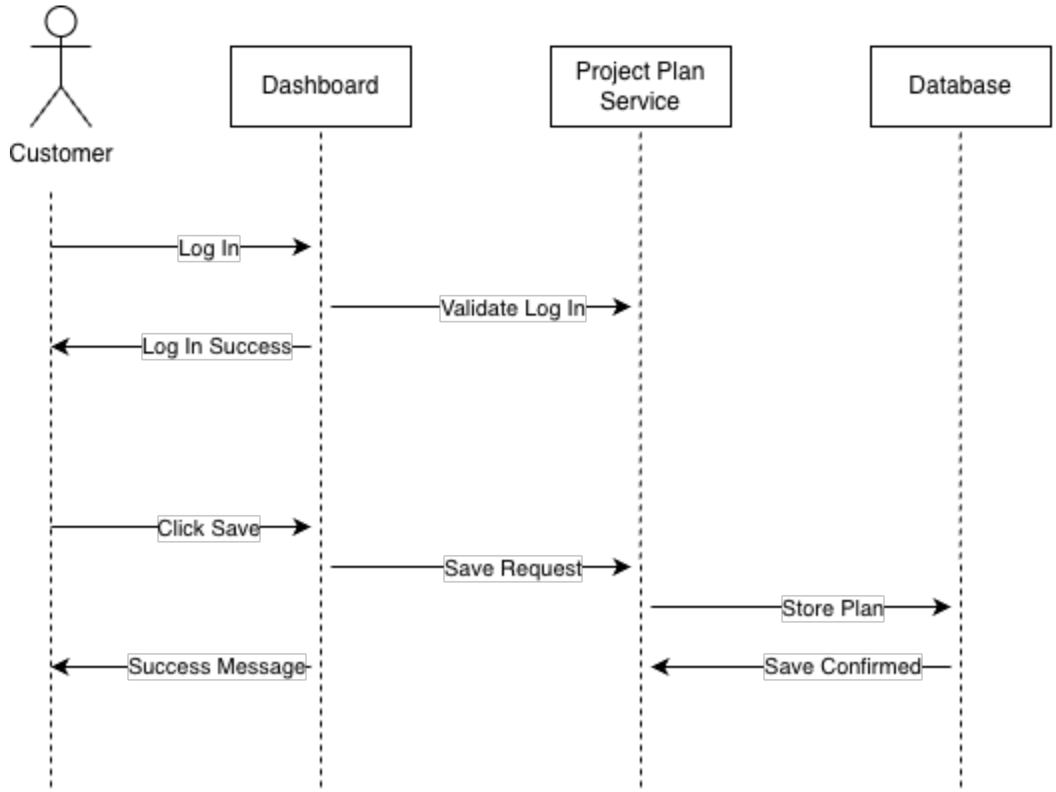


(Alen Jo, FR #1 to FR #10)



(Estrella Avila FR-21)

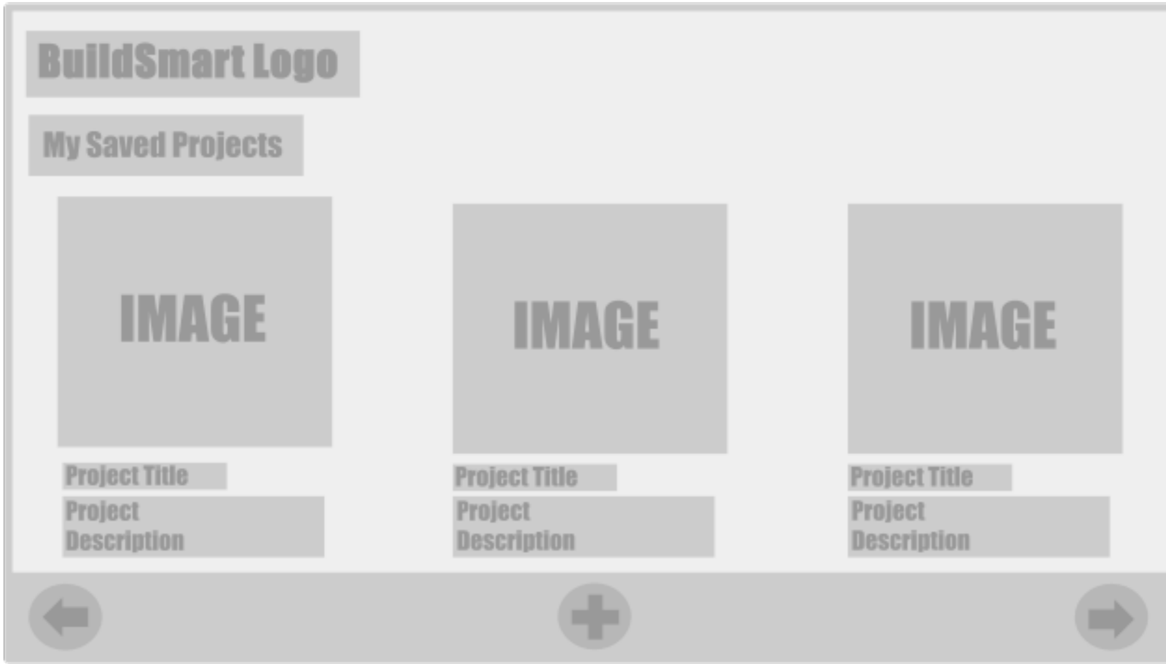
This sequence diagram models the process of a **Customer** saving an AI-generated project plan to their account dashboard. The interaction begins when the customer logs in and selects the option to save a generated plan. The request is processed by the service, stored in the database, and confirmed back to the customer.



Wire Frames

Saved Projects Dashboard (Estrella Avila, FR-21, FR-22)

This screen lets users view, save, and share project plans.



In-Store Associate Assistance Page (Estrella Avila, FR-23, FR-24)

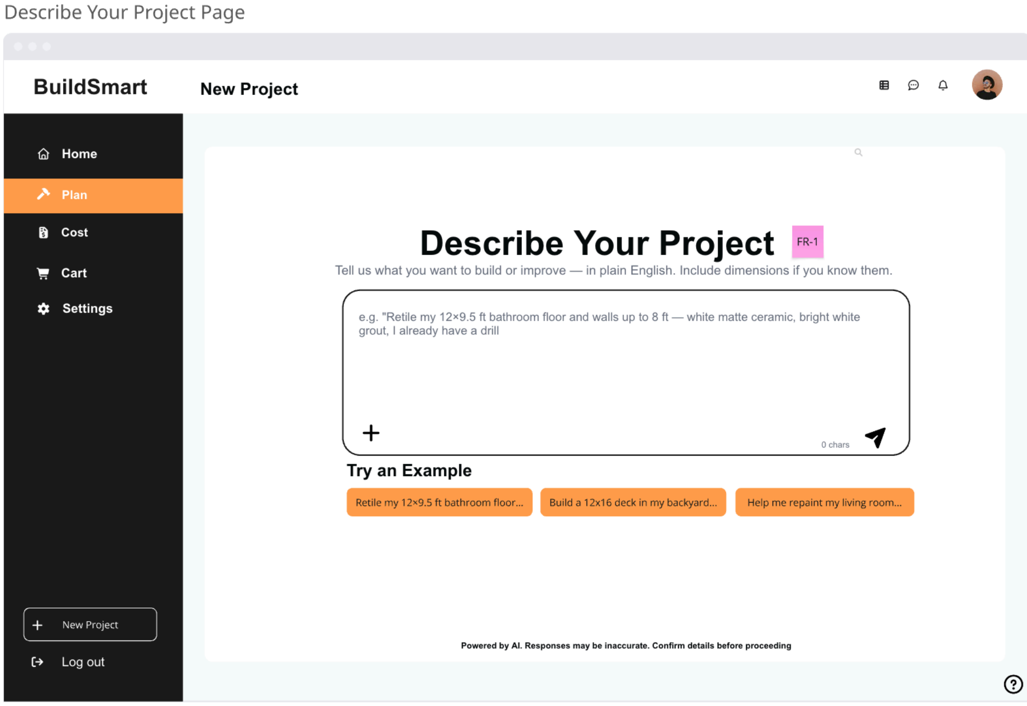
This page allows Home Depot associates to help customers find materials quickly.

Project Recommendations Page (Estrella Avila, FR-25, FR-26, FR-27, FR-28)

This screen focuses on product recommendations, promotions, and availability.

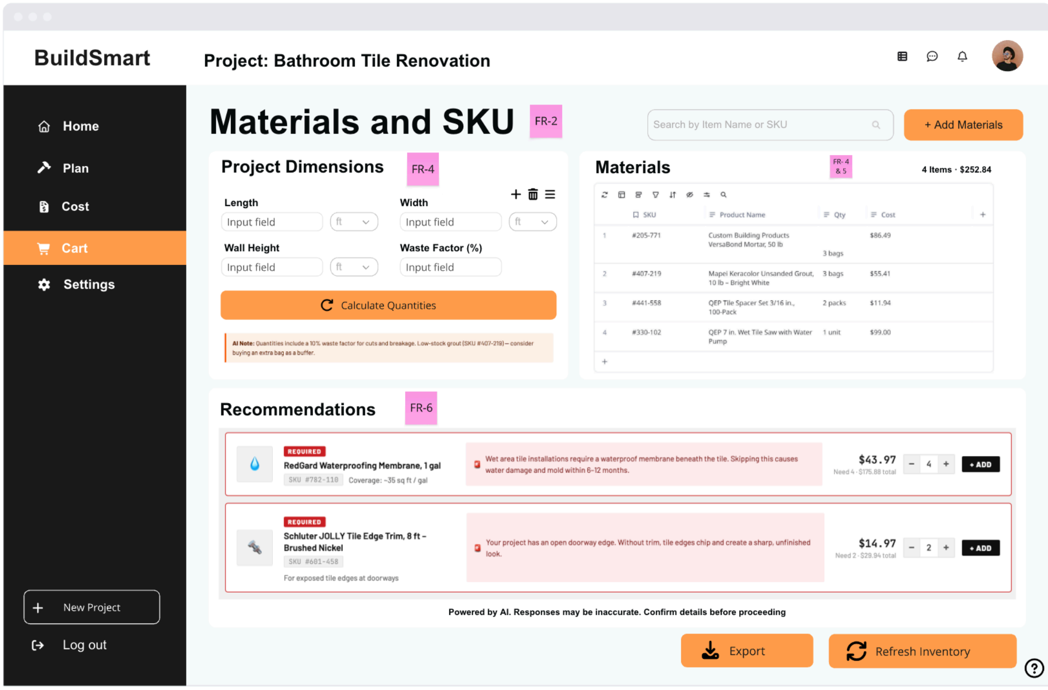
Describing Your Project Page – Alen Jo:

Describe Your Project Page



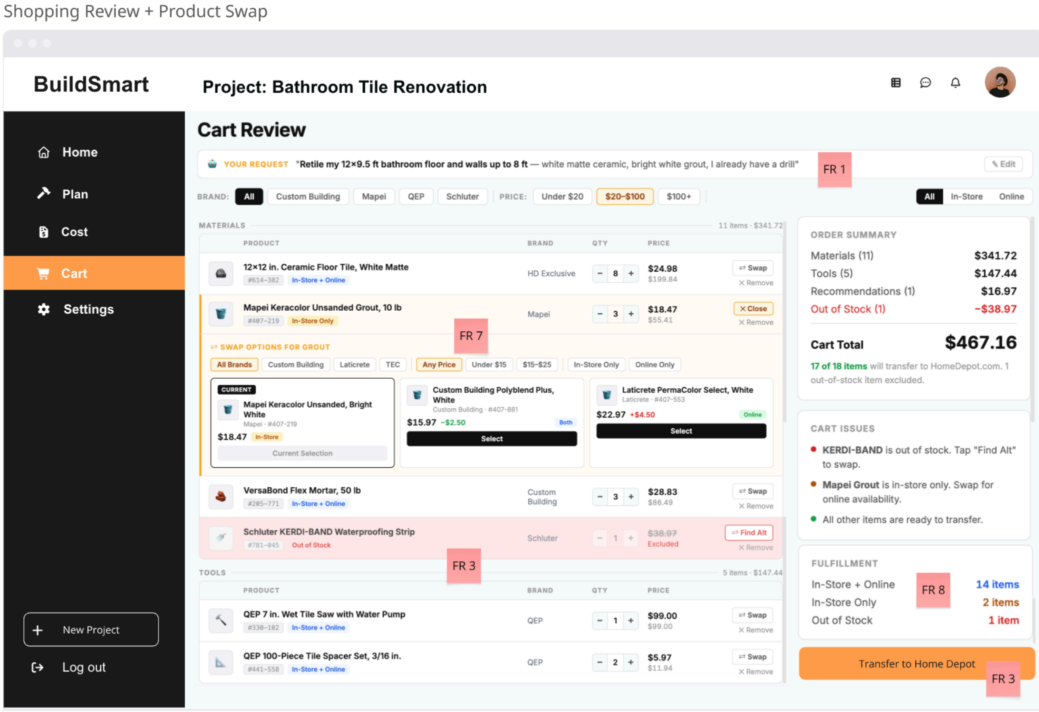
Materials and SKU Page – Alen Jo:

Materials and SKU

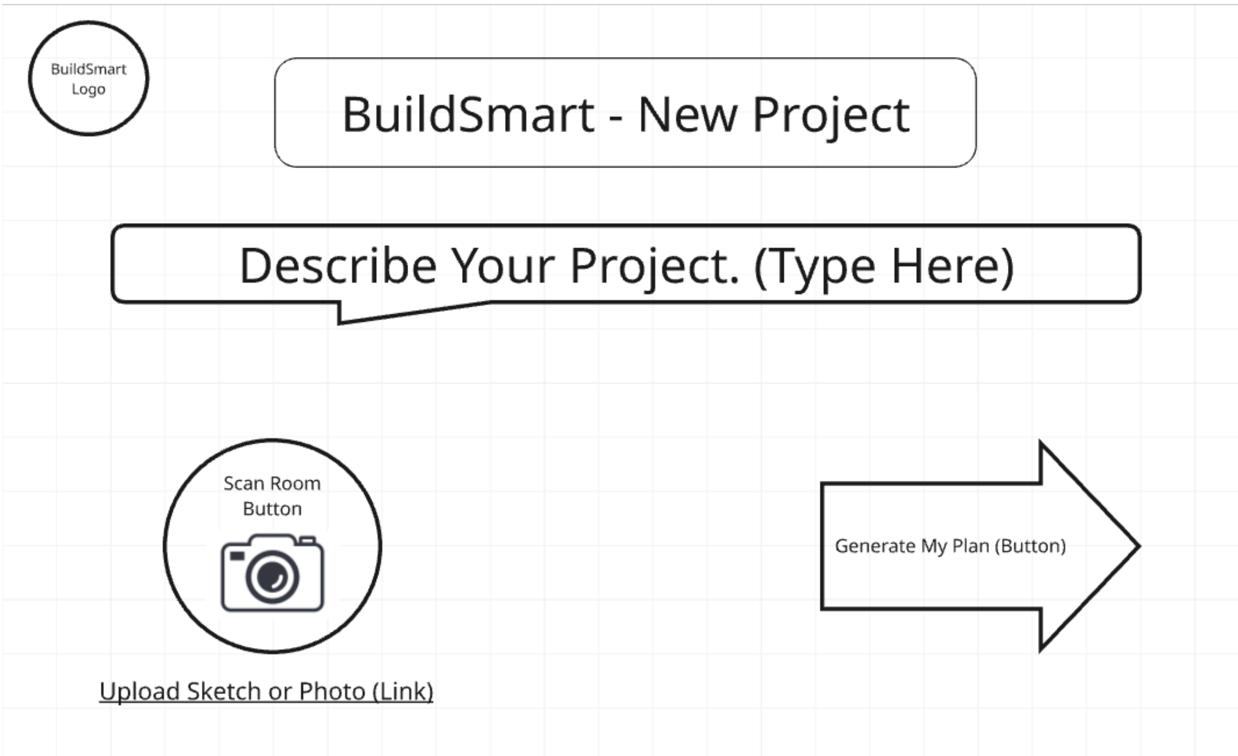


Shopping Cart Review + Product Swap Page – Alen Jo:

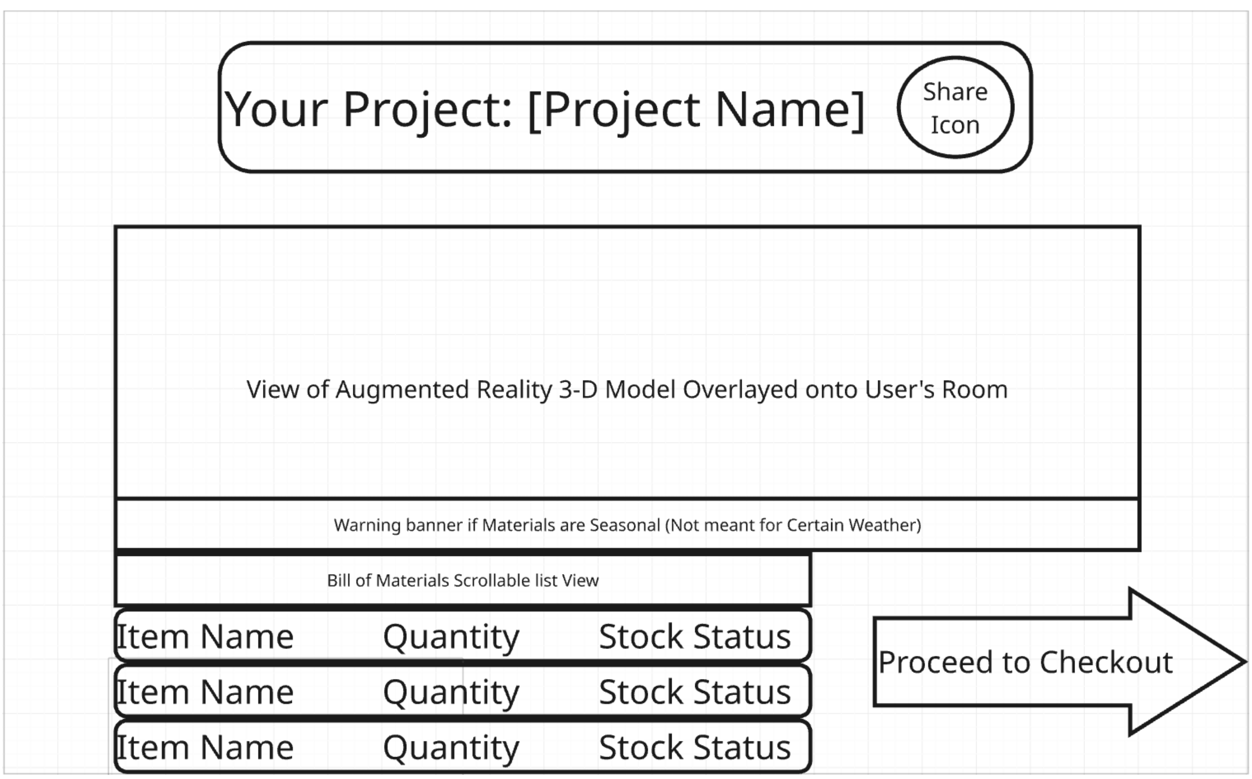
Shopping Review + Product Swap



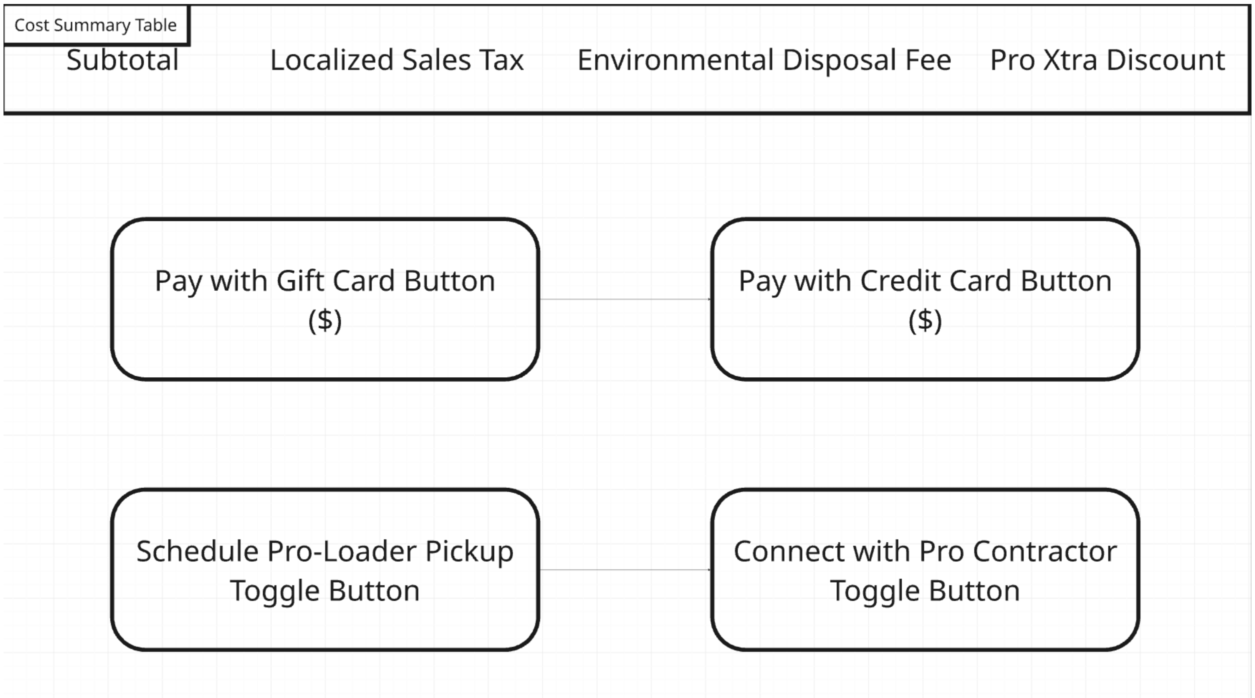
Product Entry Page – Prajit Alexander, FR-11, FR-12



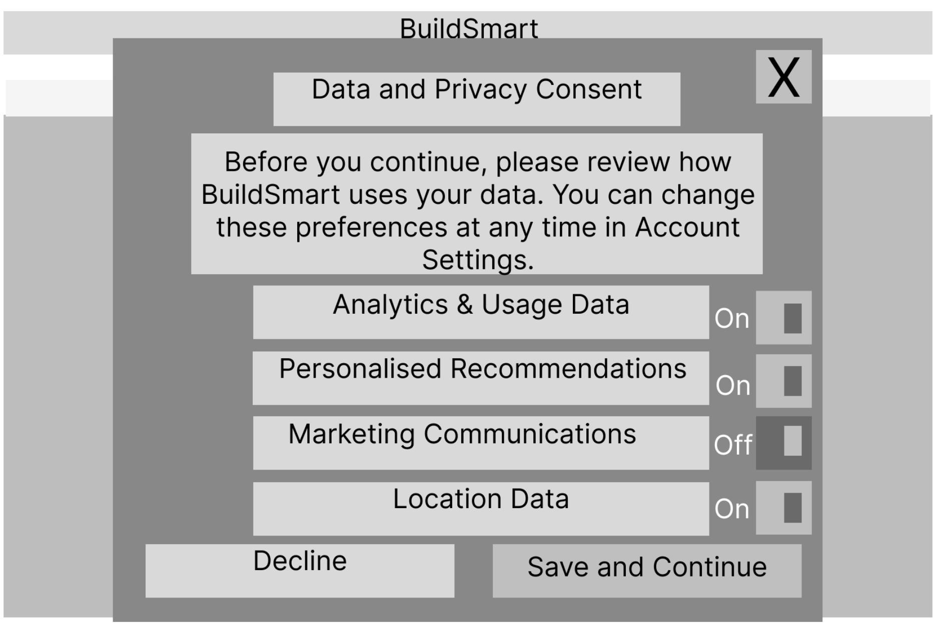
AI Plan Results Page – Prajit Alexander, FR-17, FR-19



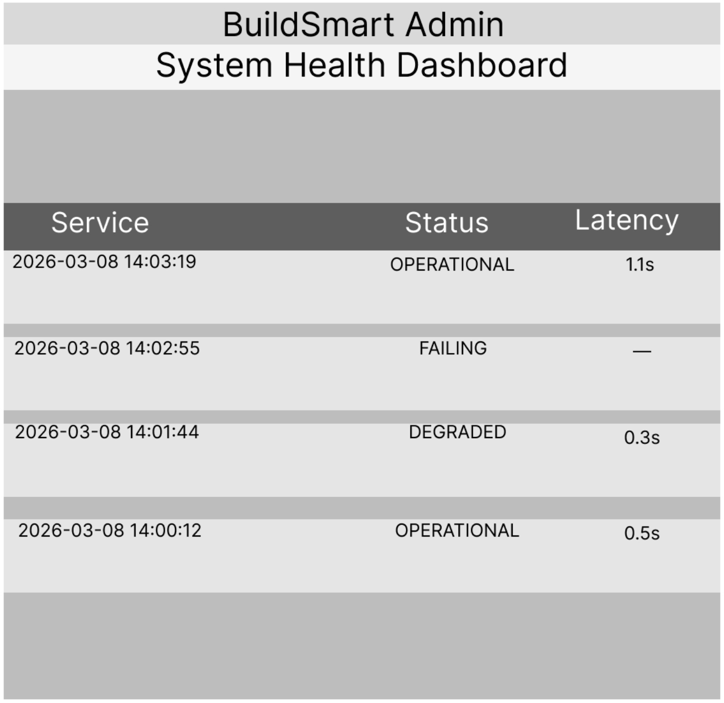
Checkout Page – Prajit Alexander, FR-13, FR-14, FR-16, FR-18, FR-20



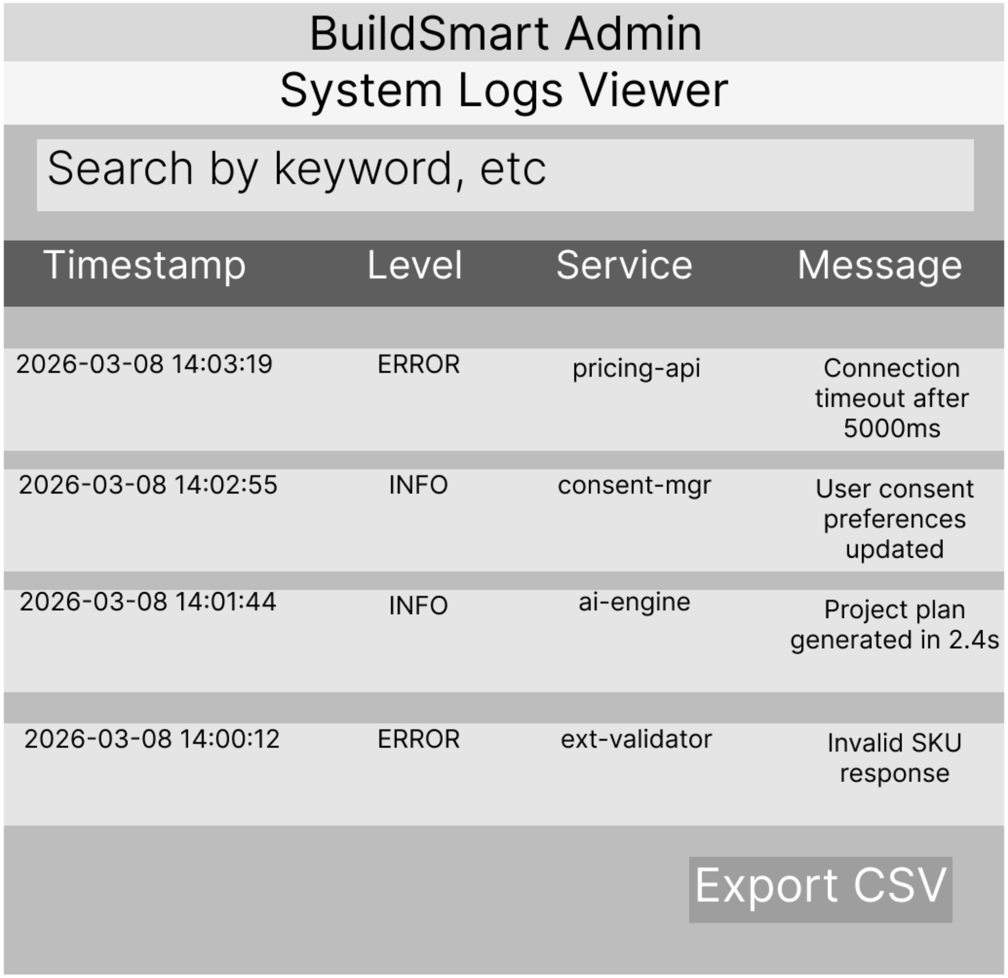
Wire Frame 10 - Privacy Consent Pop-Up – Nathan Tennyson, FR-31, FR-34



Wire Frame 11 - Admin Health Dashboard – Nathan Tennyson, FR-36, FR-37

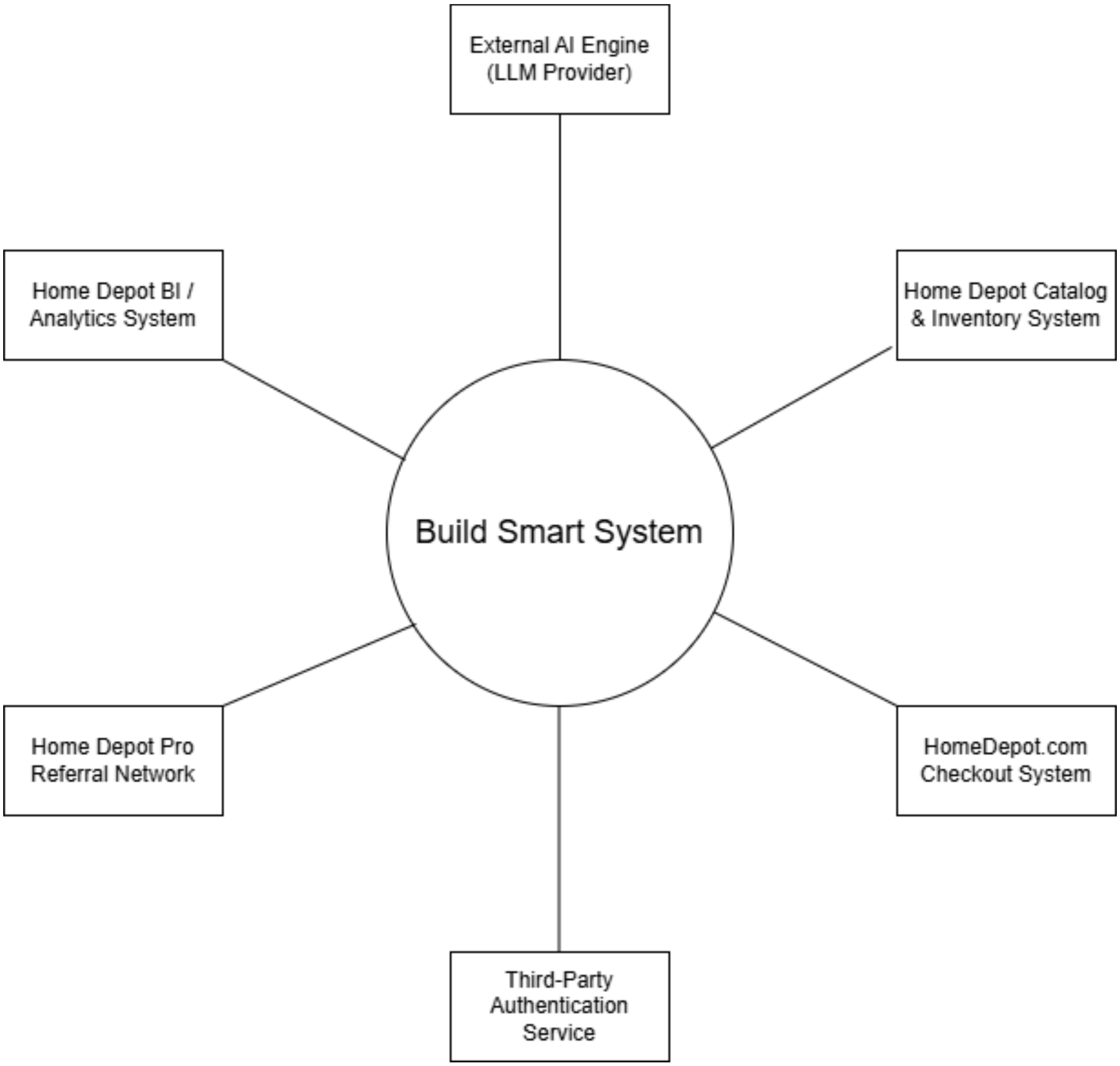


Wire Frame 12 - Admin System Logs Viewer – Nathan Tennyson, FR-38, FR-39, FR-40



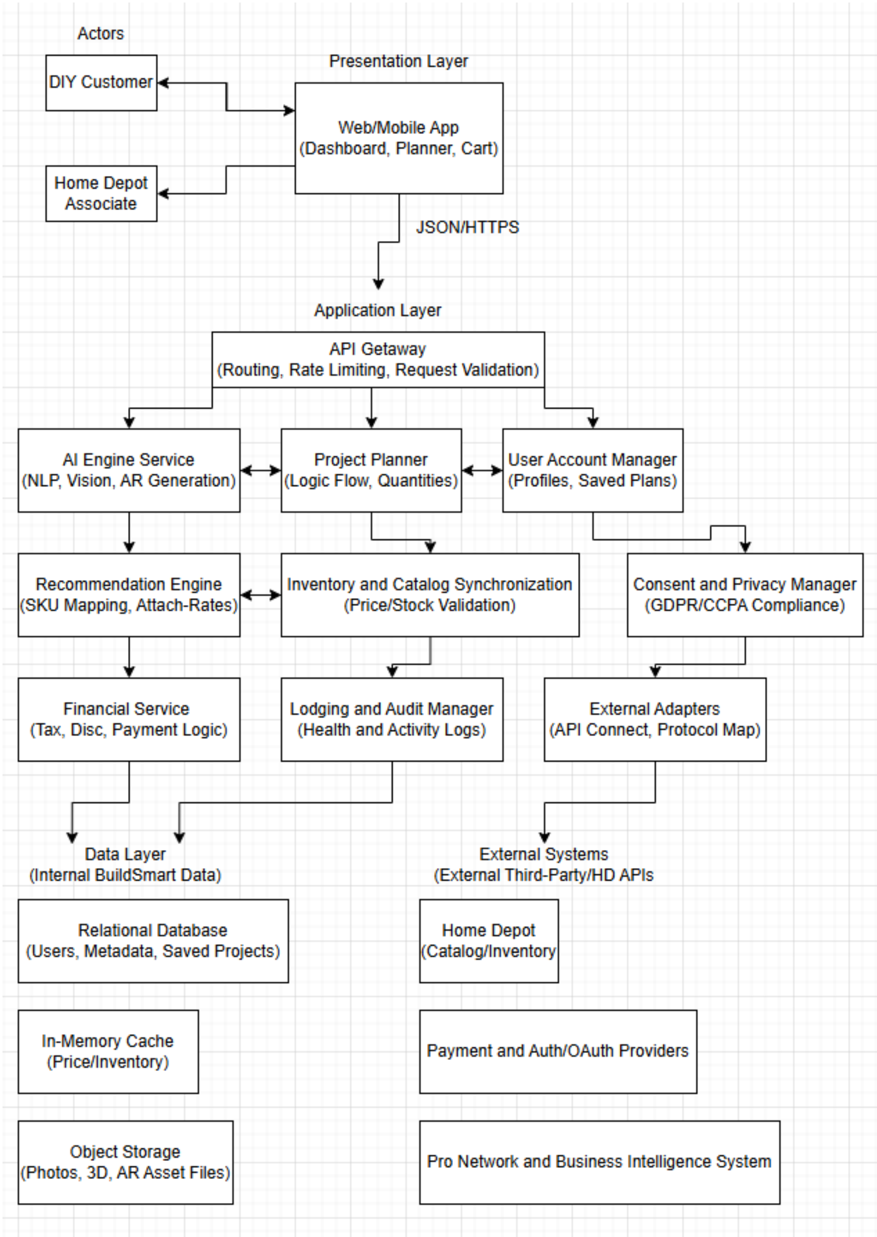
Architecture Design

Context Diagram



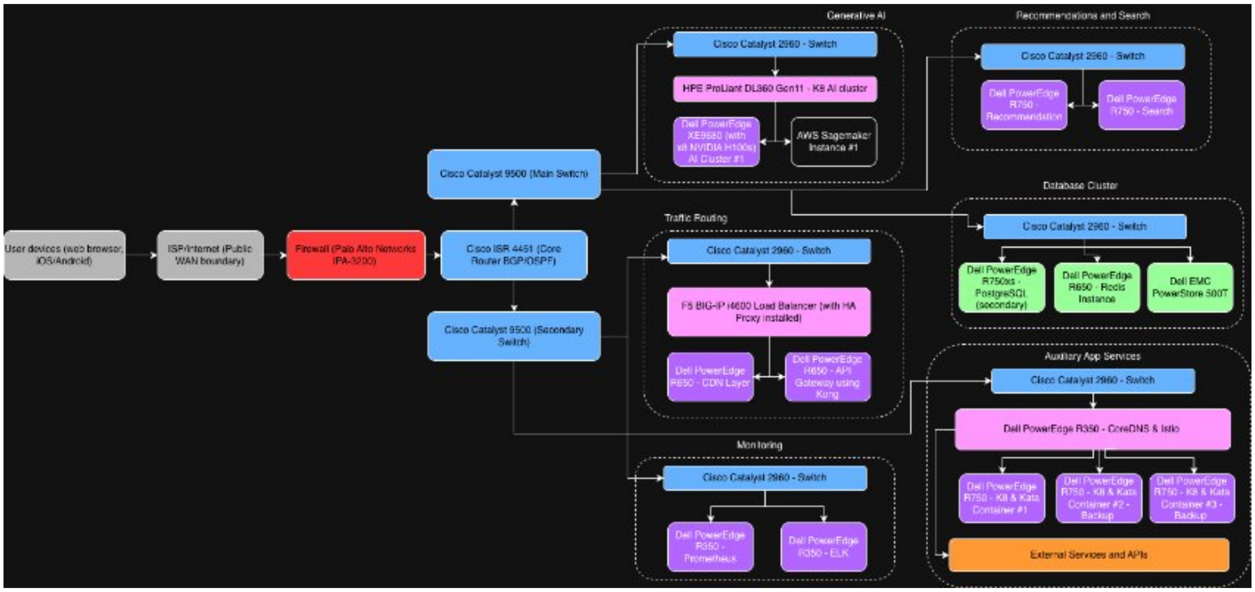
The context diagram presents the Build Smart System as the central platform and outlines its interactions with key external systems. It integrates with an AI engine (LLM provider) to generate project plans from user input, and connects to multiple Home Depot systems for product data, inventory, checkout, analytics, and professional referrals. The system also uses a third-party authentication service to manage user access and security. Overall, the diagram defines the system boundary and shows how external services support core functionality.

Systems Arch (Logical)



This Logical Architecture diagram represents the internal functional decomposition of the BuildSmart system, organized into a structured three-tier model consisting of the Presentation, Application, and Data layers. It moves beyond high-level boundaries to show the specific software modules responsible for transforming natural language project descriptions into SKU-mapped material lists. The diagram highlights the API Gateway as the entry point for all client traffic and illustrates the role of External Adapters in maintaining decoupled, resilient communication with Home Depot's external catalog and inventory systems. This view focuses on system modularity and the separation of concerns, ensuring that critical operations like localized financial calculations, secure audit logging, and privacy consent management are executed by independent and dedicated logical components.

Systems Arch (Physical)



Tech Stack

Frontend:

React Native

Backend and API:

python™

FastAPI

Database:

PostgreSQL

Caching:

redis

Cloud/Hosting:

aws

Storage:

Amazon S3

Authentication:

AWS Cognito

AI/NLP:

ChatGPT

Overview

For this project, we will use a technology stack that supports AI-drive features, real-time data handling and scalability, while still being practical to implement. The overall goal is to use technologies that work well together at supporting web and mobile applications and a backend that handles AI processing and external integrations.

Core technologies used in this system are:

- **Frontend:** React Native
- **Backend / API:** Python, FastAPI
- **Database:** PostgreSQL (+ pgvector)
- **Caching:** Redis
- **Cloud / Hosting:** AWS
- **Storage:** Amazon S3
- **Authentication:** AWS Cognito
- **AI / NLP:** LLM (ChatGPT)
- **Version Control:** Git (GitHub)
- **Containerization:** Docker
- **Monitoring / Logging:** AWS CloudWatch
- **Testing:** Pytest, Jest

Frontend

For the frontend, we are using React Native. Since this project intends to support mobile users as well as web users, React Native gives us a good balance between performance and ease of development across platforms.

Backend and API

For the backend, we are using Python with FastAPI. Python is a good choice for this project because of its compatibility with AI and NLP pipelines, which are an essential part of this system.

FastAPI is used to handle HTTP requests between the client and backend services. It allows us to build clean and efficient APIs while also supporting validation and structured responses. This fits with our architecture where the backend coordinates between the AI engine, database, and external APIs.

Database

For the database, we are using PostgreSQL as the primary relation database. This is where structred data such as user accounts, saved projects and metadata will be stored.

Additionally, we plan to use pgvector with PostgreSQL. This is important for the AI portion of the system, since it allows us to store vector embeddings and support similarity search. In the case of our Home Depot chatbot, this will allow the system to better understand user queries and provide more accurate recommendations.

Caching

For caching, we are using Redis. Redis acts as an in-memory data store that helps speed up access to frequently used data like inventory, pricing, and session information.

Cloud / Hosting

The system will be hosted on Amazon Web Services. AWS provides scalable infrastructure that supports deployment across development, testing, and production environments.

Using AWS also makes it significantly easier to integrate other services like storage, authentication, and monitoring without needing to manage everything manually.

Storage

For object storage, we are using Amazon S3. This is used to store unstructured data such as user-uploaded images, generated project files, and AR-related assets.

S3 works well here because it is scalable and reliable, especially for handling larger files that dont belong in the relation database.

Authentication

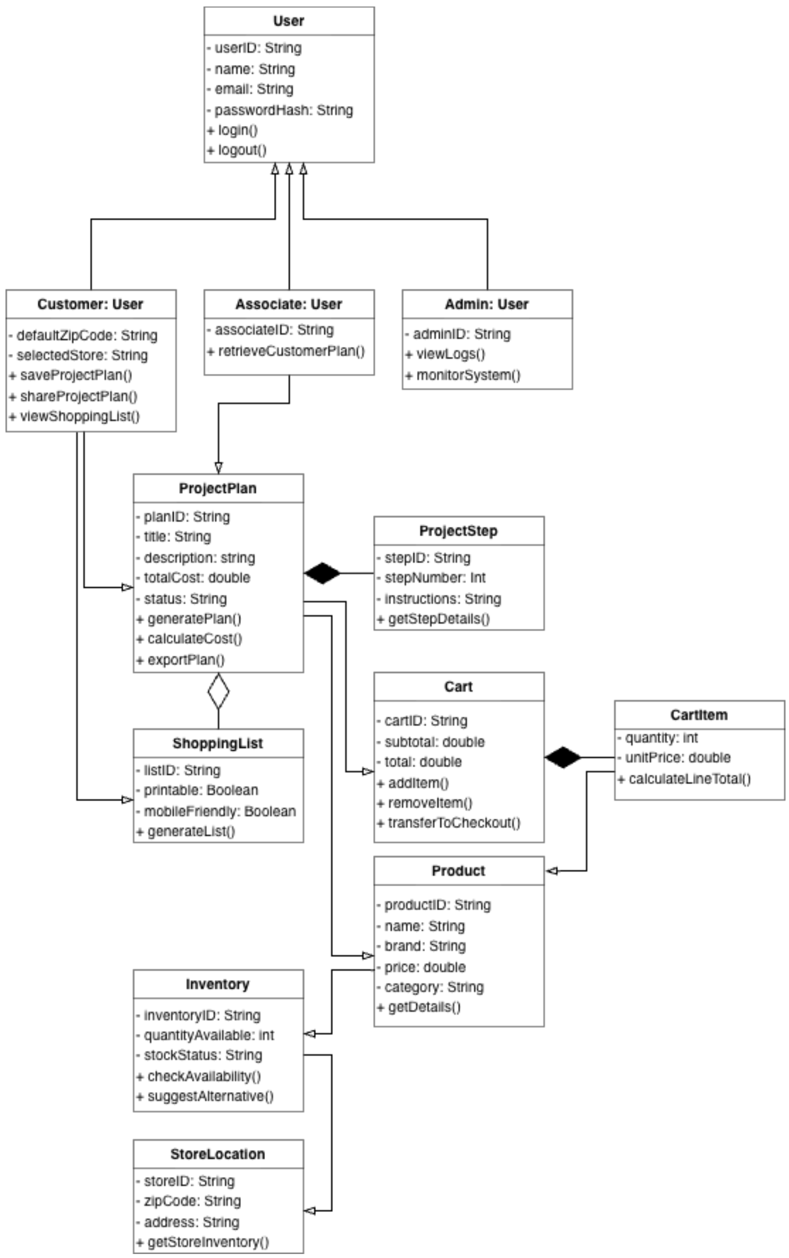
For authentication, we will use AWS Cognito. This handles user sign-up, login, and session management.

Cognito also supports OAuth 2.0 and Multi-Factor Authentication (MFA), which helps meet the systems security requirements while keeping user management relatively simple to implement.

AI / NLP

The system will incorporate a Large Language Model (LLM), such as ChatGPT, to handle natural language input and generate structured project plans.

UML Classes


User
- userID: String
- name: String
- email: String
- passwordHash: String
+ login()
+ logout()
Customer: User
- defaultZipCode: String
- selectedStore: String
+ saveProjectPlan()
+ shareProjectPlan()
+ viewShoppingList()
Associate: User
- associateID: String
+ retrieveCustomerPlan()
Admin: User
- adminID: String
+ viewLogs()
+ monitorSystem()
ProjectPlan
- planID: String
- title: String
- description: string
- totalCost: double
- status: String
+ generatePlan()
+ calculateCost()
+ exportPlan()
ProjectStep
- stepID: String
- stepNumber: Int
- instructions: String
+ getStepDetails()
Cart
- cartID: String
- subtotal: double
- total: double
+ addItem()
+ removeItem()
+ transferToCheckout()
CartItem
- quantity: int
- unitPrice: double
+ calculateLineTotal()
ShoppingList
- listID: String
- printable: Boolean
- mobileFriendly: Boolean
+ generateList()
Product
- productID: String
- name: String
- brand: String
- price: double
- category: String
+ getDetails()
Inventory
- inventoryID: String
- quantityAvailable: int
- stockStatus: String
+ checkAvailability()
+ suggestAlternative()
StoreLocation
- storeID: String
- zipCode: String
- address: String
+ getStoreInventory()

Implementation

The implementation of the BuildSmart would focus on a service-oriented design focusing on the unbroken and seamless transformation of natural language project descriptions into SKU-mapped material lists.

Development Approach

The project utilizes the Scrum Agile framework to allow for iterative progress and quick adaptation to feedback.

Sprint Structure: Development will be conducted in weekly periods (sprints) to supplement the incremental process of developing, testing, and refining the features of the system.

Team Roles:

- Nathan Tennyson: Focuses on the system health monitoring, logging, and security compliance.
- Estrella Avila: Responsible for user dashboards, plan sharing, and in-store associate interface features.
- Alen Jo: Focuses on the SKU mapping and quantity recommendation logic.
- Prajit Alexander: Responsible for NLP extraction and financial calculation requirements.

Technology Stack:

Component	**Technology**	**Justification**
Frontend	React Native	Provides a high-performance experience for both web and mobile users across platforms.
Backend and API	Python & FastAPI	Python is essential for AI/NLP compatibility, while FastAPI ensures efficient HTTP request handling and structured responses.
Database	PostgreSQL	Acts as the main relational store for metadata; includes pgvector to enable

		similarity searches for AI recommendations.
Caching	Redis	Utilized as an in-memory data store to accelerate access to frequently updated inventory, session, and pricing data.
Cloud/Hosting	AWS	Provides reliable infrastructure to support deployment across development, testing, and production environments.
Storage	Amazon S3	Securely stores unstructured data, including user-uploaded photos, 3D models, and AR-related assets.
Authentication	AWS Cognito	Manages secure user sign-up and login, meeting security standards of MFA and OAuth 2.0 support.
AI/NLP	ChatGPT	Integrated as the Large Language Model (LLM) to transform natural language user input into structured project plans.

Development Environment:

BuildSmart utilizes a cloud-native infrastructure hosted on Amazon Web Services (AWS) to maintain consistency across the software lifecycle.

- Development Environment: Local development is performed using Python and React Native, with initial unit testing conducted via Pytest and Jest.
- Testing Environment: This environment is essentially a mirror of the production setup to evaluate the system with functional and non-functional requirements.
- Production Environment: The live environment is tuned for high availability, implementing AWS CloudWatch for real-time monitoring and the logging of system events.

Configuration Management

The team follows a structured configuration protocol to ensure code integrity and streamline the integration of AI-driven features.

- Version Control: Git is the primary version control system, with GitHub serving as the central holder for code availability.
- Branching Strategy: All updates follow semantic versioning to maintain order and ensure that every change is trackable.
- Integration Process: Changes must pass existing and new test cases before deployment to ensure no service disruptions occur.
- Containerization: Docker is utilized to package the application and its dependencies, ensuring that the backend, which blends the AI engine and catalog services, runs consistently across all environments.

Coding and Integration Plan

The system is organized into a three-tier model to ensure the platform is resilient to changes in Home Depot's product catalog.

- Presentation Layer: A cross-platform interface built with React Native that allows users to interact with the AI Project Chat and view 3D AR models.
- Application Layer: The backend, powered by FastAPI, acts as the coordinator between the ChatGPT AI engine, the internal project planner, and external catalog APIs.
- Data Layer: Manages internal PostgreSQL data, Amazon S3 object storage for photos, and Redis for caching real-time inventory levels.

Execution Phases:

- AI Integration: Setting up Natural Language Processing (NLP) to extract the project scope and materials from descriptions.
- Data Synchronization: Adapters sync pricing and SKU data from the Home Depot catalog API at intervals not exceeding 15 minutes.
- Security & Encryption: Hardening the system with AES-256 encryption for sensitive data and Multi-Factor Authentication (MFA) for administrative access.

- Continuous Testing: Verifying end-to-end paths—from natural language query to the transfer of an itemized cart—to ensure all functional requirements are met.

Prototype (MVP) Implementation

Prototype Development

The BuildSmart prototype represents a functional Minimum Viable Product (MVP) that demonstrates the core system workflow, from initial project input to material planning. The prototype allows users to enter a home improvement project description in natural language, which is then processed by an AI component to generate a structured project plan.

The system produces a Bill of Materials (BOM) that includes mapped products, estimated quantities, and cost information. Users can review this information in a cart-style interface and make basic adjustments, such as selecting or swapping products. The prototype also supports saving and revisiting project plans through a user account dashboard, along with generating a simple shopping list for added convenience.

Some advanced features such as augmented reality visualization, full checkout functionality for exporting the cart directly to HomeDepot.com, real-time pricing integration, and more advanced product filtering were intentionally simplified or only partially implemented. These limitations were mainly due to technical constraints and the need to keep the scope focused on core functionality.

Overall, the MVP focuses on demonstrating the key value of the system by turning a user's high-level project idea into a clear, structured plan with relevant materials and cost estimates. At the same time, it provides a solid foundation for future improvements that can expand the system to fully meet all requirements.

Code Contribution

Team Member	Contribution
Nathan Tennyson	Implemented user authentication and account management for the MVP, including login and account creation using a unified backend endpoint. Developed frontend pages for Login, Create Account, and Account dashboard, and implemented localStorage-based session handling. Added project persistence by linking saved projects to user accounts in the backend, allowing users to revisit and reopen projects.

	Updated routing, sidebar, and navbar navigation, and implemented logout functionality.
Estrella Avila	Implemented the saved projects dashboard for BuildSmart which fetches the user's saved project plans from the backend, displays each project as a card, and allows users to open a saved plan by passing the plan data through router state to the result page. Cards also have an edit and delete controls for managing saved projects, along with icon-based designs for a cleaner dashboard layout.
Alen Jo	Architected and implemented the full authentication system from scratch for components like the backend endpoints and frontend scaffolding. Helped setup the PostgreSQL database and host it via cloud through Neon PostgreSQL. I also created out the Docker files needed to containerize the backend service for our project, where I needed to make sure I have our automated browser running in background to mark Home Depot GraphQL requests as valid. Through that, I did investigative programming into how the Home Depot GraphQL API worked and was successfully able to enrich our natural query pipeline with product recommendations, in-stock items, and zip code for any store location. Moreover, I went ahead and built out the BI export pipeline that takes in user chosen products and aggregates their SKUs to create views that are exportable to services, such as Tableau or as Parquet for more advanced Big Data Management services. I also managed to write up some CI/CD tests for the frontend and backend to verify and assert that our application is in a healthy state. I also made sure to host our application via a local server through Cloudflare Tunnels and various trade-offs in using a cheap VPC vs. a hosting platform, although it was not worth the cost in procuring those things. In short, I did a lot of my requirements and ensured that our prototype is working and operational to present. Finally, I did code reviews for the rest of my team to ensure that no one's change breaks and followed proper Git repository strategies.
Prajit Alexander	Focused on the localized financial and logistics modules to ensure the platform's utility at a state level. Implemented addition to the cost-calculation engine, allowing it to automate the inclusion of regional sales taxes based on specific user zip codes to provide accurate final costs. Integrated real-time scheduling, enabling users to view real-time availability and

	understand how to secure Pro-Loader assistance for in-store pickups.

Requirements Alignment

Requirement ID	Requirement Description	Implemented	Notes	Implemented By
FR-1	The system shall allow users to input a home improvement project description in natural language and receive a structured project plan in response.	Yes	Users can input a project description and receive a structured plan generated by the AI component	Alen
FR-2	The system shall map every step of a project plan to specific, purchasable Home Depot SKUs with real-time pricing.	Partial	Product mapping is implemented using a predefined or mock dataset rather than real-time Home Depot API integration	Alen
FR-3	The system shall generate a complete, itemized cart that users can review and transfer directly to HomeDepot.com for checkout.	Partial	Itemized cart is generated and displayed as a CSV, but not possible to export it via Home Depot due to constraints with exporting a shopping cart snapshot back to user	Alen

FR-4	The system shall provide quantity recommendations for all materials based on user-provided dimensions or project scope.	Yes	Basic or estimated quantities are provided with precise calculations from natural query.	Alen
FR-5	The system shall display estimated total project cost, broken down by materials, tools, and optional add-ons.	Yes	Total cost is calculated and displayed based on selected or generated items	Alen
FR-7	The system shall allow users to filter, or swap recommended products by brand, price range, or availability (in-store vs. online).	Partial	Users can perform basic product swapping or selection, but advanced filtering options are limited for things based on brand, availability, or other fine-grained filters	Alen
FR-9	The system shall provide step-by-step project guidance alongside each recommended product to improve DIY confidence and reduce misuse-related returns.	Yes	Generated plan includes step-by-step guidance associated with the project	Alen
FR-11	System shall use Natural Language Processing to extract project scope, materials,	Yes	NLP functionality is implemented using an LLM to interpret user	Alen

	and intent from free-text user descriptions.		input and generate structured output	
FR-13	The system shall calculate final project costs including local sales tax and environmental disposal fees based on the user’s specific zip code.	Partial	Local Sales tax is implemented, but environmental disposal fees implementation is not.	Prajit
FR-14	The system shall provide available delivery slots for bulk materials and allow users to schedule "Pro-Loader" assistance for in-store pickups	Partial	Users can look at available pro-loader time slots.	Prajit
FR-21	The system shall allow users to save and revisit previously generated project plans through a user account dashboard.	Yes	Users can save and view previously generated project plans	Estrella and Nathan

Wireframe Alignment

Wireframe Name	**Related Requirements**	**Implemented**	**Notes**	**Implemented By**
Describing Your Project Page	FR-1, FR-11	Yes	The implemented project input screen matches the wireframe’s core purpose by allowing users to	Alen

			enter a project description in natural language	
AI Plan Results Page	FR-1, FR-9, FR-11	Yes	The implemented results screen displays the generated structured project plan and step-by-step guidance in a format consistent with the wireframe	Alen
Materials and SKU Page	FR-2, FR-4	Yes	The implemented page displays mapped products and estimated quantities, with actual product data integrated through the Home Depot API service	Alen
Shopping Cart Review + Product Swap Page	FR-3, FR-5, FR-7, FR-13, FR-14 FR-24	Partial	The implemented cart screen shows selected items, cost breakdown, pro-loader schedule choices, shopping list information, and basic product replacement (in-lined per product), but advanced filtering and direct	Prajit and Alen

			HomeDepot.com transfer is limited	
Saved Projects Dashboard	FR-21, FR-22	Partial	The implemented dashboard allows users to save and revisit project plans, while sharing functionality is included in a simplified form	Estrella and Nathan

Feature Mapping

Feature	Requirement(s)	Description
Project Input	FR-1, FR-11	Users enter a home improvement project description using natural language input, which is processed by an AI model to generate a structured plan
AI Project Plan Generation	FR-1, FR-9	System generates a structured, step-by-step project plan with guidance for completing the project
Materials & SKU Mapping	FR-2, FR-4	System maps project steps to materials, provides estimated quantities, and links to purchasable SKUs (mock dataset in MVP)
Cost Estimation Engine	FR-5, FR-13	Calculates and displays total project cost, including material breakdown and local sales tax based on zip code
Shopping Cart & Product Review	FR-3, FR-7	Displays itemized cart with ability to review, select, or swap products (limited filtering in MVP)
User Authentication & Account Management	FR-21, FR-22	Supports account creation, login, logout, and session handling for personalized project storage
Product Selection & Replacement	FR-7, FR-28	Allows users to swap recommended products with alternatives (basic functionality in MVP)
Scheduling & Pro-Loader Integration	FR-14	Users can view available time slots for in-store pickup assistance (Pro-Loader)

Code Availability

https://github.com/joalen/BuildSmart

Software Testing

Testing Strategy

Like with testing, we only finish testing once our test cases pass as defined for Home Depot. During development, TAJA Co will write up unit tests to ensure that the code written behaves what it is expected to. After development, TAJA Co will create and run integration and regression tests to make sure that this project planning application works with all components seamlessly and to catch early bugs. After, TAJA Co will conduct a User Acceptance Test (UAT) to ensure that Home Depot is satisfied with the product and that all expected behaviors are working.

Testing Technologies and Methodologies

1.1 - Unit Testing:

TAJA Co will deploy automated test frameworks that apply to any discrete function, method, and class applicable to the codebase. Each unit will be tested in isolation through these frameworks to verify the correctness of expected behavior.

1.2 - Integration Testing:

TAJA Co will run integration testing for all frontend, backend, and database layers. This ensures that we have all operations running smoothly in an interconnected way. TAJA Co will test frontend interfaces and analyze data flows from client to server to make sure that the expected functionality works and is secure for each client. There's also component testing that TAJA Co will do to catch errors not found in unit testing and comes after combining multiple individual modules together to work on a problem.

1.3 - System Testing:

TAJA Co will then evaluate the entire application against Home Depot's requirements (functional and non-functional) through a mirror of the production environment called staging. Test cases come from Home Depot's requirements and expected sequence diagrams to ensure that documented user interaction paths are followed. There's also end-to-end testing to make sure that everything is working correctly from point A, a user asking a particular query for their project, to point B, receiving the BOM with product recommendation that can transfer to Home Depot's website.

1.4 - Release Testing:

TAJA Co will then hand-off product to the Quality Assurance team that uses a black-box approach to derive potential tests to run on this product. TAJA Co's QA Team will use cases derived from Home Depot as opposed to the internal implementations of the system. The purpose is to make sure that TAJA Co has met the contracted requirements as stated by Home Depot before being handed off to customer for final inspection.

1.5 - User Acceptance Testing:

Finally, TAJA Co will hand off the customer's product to representatives of Home Depot. Representatives will ensure that the system meets business and operational requirements before it goes live to the public. Home Depot representatives will "mock" customer journeys, such as planning a project, configuring the cart, fine tuning product recommendations and project, and reviewing project costs. Throughout this "mocked" customer journey test, representatives take note of observations noticed of the product. The customer would need to sign off on this UAT Acceptance if ultimately Home Depot is satisfied. Afterwards, TAJA Co can begin deployment.

1.6 - Testing Technologies and Methodologies

For frontend, TAJA Co will employ the JavaScript Jest Framework and Playwright to ensure that end-to-end testing is successful for actions and that all asserted actions are correct. Backend will utilize Pytest for hosting unit, regression, and integration tests across modules plus Postman to check if API responses follow proper, expected payloads needed for BuildSmart to work.

TAJA Co will utilize Amazon AWS CloudWatch to monitor BuildSmart's real-time issues and logging to catch errors along with JMeter for performance and scalability testing as needed necessary.

As for AI response quality, we will employ the use of Chat response quality tools that ingest in user conversations and assistant responses to see weak spots in responses and potential points of latency. This'll help TAJA Co determine appropriate LLM vendors to take in and budget control as stated by Home Depot.

Test Cases

Functional Requirements Test Cases

Test Case #	Related Requirement	Test Description	Test Steps	Expected Result
TC-01 (Alen)	FR-01	Verify that a clear, well-written project description produces a complete structured plan	1. Navigate to project input screen. 2. Type anything (e.g - “I want to repaint my living room walls.”) 3. Click submit	System produces a structured plan with labeled sections like: - Overview - Materials - Tools - Step-by-Step instructions
TC-02 (Alen)	FR-02	Verify that every material and tool in a generated plan links to a specific Home Depot SKU and product with current price	1. #TC-01 steps 2. Review the materials list generated from TC-01 3. Make sure that the displayed SKUs align with what Home Depot has	Every item in the plan generated should cross-reference to that exact item found in Home Depot’s website
TC-03 (Alen)	FR-03	Verify that system builds complete itemized cart from project plan and transfers it appropriately to homedepot.comfor final checkout	1. #TC-01 2. #TC-02 3. Now confirm each item shows the proper SKU, unit price, quantity, and name. 4. Click “transfer to Home Depot” 5. Inspect the cart at homedepot.com	Cart populated with the proper items as listed from generated plan and after user goes to homedepot.com, the cart should be filled with those items as exactly stated.
TC-04 (Alen)	FR-04	Verify system calculates and recommends accurate material quantities when user provides room	1. #TC-01 2. Go to the materials and SKU page 3. Add specifications	System recommends proper quantity of materials based off giving room specifications;

		specifications (i.e - dimensions)	such as room length, width, volume, etc. 4. Review recommended quantities based off room specs through some manual verification	need to manually check if calculation are correct
TC-05 (Alen)	FR-05	Verify system gives a grand summary of cost and does an itemized breakdown of it by material, tool, and optional add-ons.	1. #TC-01 2. Go to the shopping cart + product swap screen 3. Find the cost section and see that the breakdown is correct after sum and inclusion of state tax (if applicable)	Valid total cost that accounted for proper tax and subtotals from the individual items.
TC-06 (Alen)	FR-06	Verify system provides relevant product recommendations along with primary materials to improve basket completeness	1. TC #01 2. Scroll to see the recommendations section that came after generating plan 3. Verify suggestions are contextually relevant to drywall work 4. ‘Add to cart’ on one recommendation 5. Confirm added item doesn’t duplicate any item already in primary materials list	At least one relevant item appears that shows its relevant SKU and price. Item should be added successful and must not duplicate.

TC-07 (Alen)	FR-07	Verify users can filter products by brand, price range, and availability. Verify ability to swap recommended product for an alternative SKU	1. TC #01 2. Open the shopping cart + product swap page 3. Use the navbar to filter product by name or price range or nearest store 4. Swap any item and make sure that new item is reflected in the shopping cart 5. Old item should also be removed from the screen	Filter should hide all products that do not match the query, such as product name, availability, or price range. If product is swapped, new item should replace the old one in the shopping cart screen and has proper fields like its SKU, name, unit price, and quantity.
TC-08 (Alen)	FR-08	Verify system checks and displays real-time product availability through zip code and location. It should reflect statuses such as in-stock/out-of-stock or in-person/online only	1. #TC-01 2. Enter zip code plus the product name of interest in the Materials and SKU page for finding products 3. Verify that we've filtered results and see availability statuses: in-stock or out-of-stock 4. Also verify if it is an online-only order or in-person order	Each SKU/product should show their availability statuses. Out-of-stock should clearly be red and cannot be added to order.
TC-09 (Alen)	FR-09	Verify that the system provides ordered product-linked, safety-aware instructions for project	1. #TC-01 2. Now, within the product swap page, click on the AI modal and add in "replacing an electrical outlet"	Steps appear in logical order without gaps. Each product-dependent step will reference relevant SKUs inline. Steps must

			3. Review that the steps are proper and comply with safety code and standards as appropriate 4. Make sure that steps also tie to the SKU in question	adhere to building codes and regulations.
TC-10 (Alen)	FR-10	Verify SKU, quantity, and frequency data can be exportable to Home Depot's Business Intelligence provider and must strip all PII	1. Repeat #TC-01 x3 2. Wait for some time (like five minutes) 3. Go to Home Depot's BI system 4. Locate a particular SKU from the generated plans in #1 5. Inspect all fields and logs to see if PII is not present	There should be records for a particular SKU of interest in the BI system, and the record arrives in appropriate time. Record should not contain PII.
TC-11 (Prajit)	FR-11	Verify that system correctly takes materials and dimensions from a free-text prompt.	User navigates to new project entry screen; types wanted project and generates plan.	System generates a Bill of Materials with needed materials and correctly calculated space.
TC-12 (Prajit)	FR-12	System identifies structural obstacles from drawn sketches or photos.	Select to upload photo or sketch and then upload with a visible obstruction in background	System flags obstruction as a constraint in the resultant project summary.
TC-13 (Prajit)	FR-13	Validate that total costs reflect taxes for specific regions and environments.	Generate a plan for a specific material and enter a zip code.	Final total cost should include exact sales tax and any applied local environmental

				fees related to the acquisition of material.
TC-14 (Prajit)	FR-14	Verify the availability of Pro-Loader slots for in-store pickup.	Select the “Schedule Pro-Loader Pickup” toggle on the Checkout page.	The system shows a calendar with all available 20-minute windows of assistance.
TC-15 (Prajit)	FR-15	Confirm the ability to set up automated maintenance subscriptions.	Complete a build project, then select an annual maintenance for the upkeep of the built project.	The current users profile shows an active Subscription for maintenance.
TC-16 (Prajit)	FR-16	Test the link between the Bill of Materials and an external contractor.	Click the “Connect with Pro Contractor” toggle on the Checkout page.	A connection is generated for local Home Depot authorized contractors with the materials and measurements.
TC-17 (Prajit)	FR-17	Validate that climate-sensitive materials trigger the warning.	Input low-thermal conductivity materials in a month with forecasted freezing temperatures.	A warning banner appears to state that the scanned materials are not recommended and asks the user to reschedule or use alternate materials.
TC-18 (Prajit)	FR-18	Verify automatic discounts are applied for any Pro Xtra members.	Log into an account with Pro Xtra and add large amounts of any material to the cart.	Membership discounts are applied automatically and subtract from the total.
TC-19 (Prajit)	FR-19	Ensure the Augmented Reality model can correctly project onto a physical space.	Open the AR tool on a small device (mobile) and point the camera at an open space.	The correctly scaled 3D model of your project appears onto the floor relative to the room.

TC-20 (Prajit)	FR-20	Test the system's ability to process two separate payment types.	Traverse to the Checkout page, enter a credit card for one half and gift card for the second half.	Both payments are processed without any error and the order is marked as "Paid."
TC-21 (Estrella)	FR-21	Verify that users can save project plans.	User generates project plans and selects save project.	Project plan is stored in user dashboard.
TC-22 (Estrella)	FR-22	Verify users can share saved project plans.	User selects share project.	System generates shareable link or file.
TC-23 (Estrella)	FR-23	Verify associates can retrieve customer project plans.	Associate searches project ID in the system.	System retrieves and displays project plans.
TC-24 (Estrella)	FR-24	Verify shopping list generation for in-store navigation.	User views on recommended materials.	System displays applicable promotions or discounts.
TC-25 (Estrella)	FR-25	Verify trending projects recommendations.	User opens recommendations page.	System displays trending project suggestions.
TC-26 (Estrella)	FR-26	Verify promotional offers display.	User views on recommended materials.	System displays applicable promotions or discounts.
TC-27 (Estrella)	FR-27	Verify that in-stock products are prioritized.	User selects store location	System prioritizes items currently in stock
TC-28 (Estrella)	FR-28	Verify alternative product suggestions.	The selected item is out of stock.	Systems recommend substitute products.
TC-29 (Estrella)	FR-29	Verify anonymized usage data collection.	Users generate project plans.	System records anonymized usage session data.
TC-30 (Estrella)	FR-30	Verify analytics report generation.	The analyst requests a report.	System generates summary of

				popular projects/materials.
TC-31 (Nathan)	FR-31	Verify that the system requires user consent for PII collection during account registration.	Create a new account and proceed through the registration flow.	An in-app consent prompt appears before any PII is collected, and registration cannot be completed without addressing it.
TC-32 (Nathan)	FR-32	Verify that the system flags or deletes user data after a defined inactivity period.	Create a project plan and saved materials list, then simulate the configured inactivity period passing	The system automatically flags or removes the project data, materials list, and account information per the retention policy.
TC-33 (Nathan)	FR-33	Verify that users can view, export, and delete their personal data through the Privacy center.	Log in, navigate to the privacy center, and submit a personal data deletion request.	The system displays personal data, allows export, and confirms the deletion request will be completed within 30 days.
TC-34 (Nathan)	FR-34	Verify that users can opt out of personalized recommendations and behavioral tracking from account settings.	Log in, open account settings, and disable personalized product recommendations and behavioral data tracking.	The system saves the opt-out preferences and stops personalized recommendations and tracking for that account.
TC-35 (Nathan)	FR-35	Verify that the system generates secure audit logs for user data access, changes, and deletions.	Perform data access, an account update, and a deletion request on a user account	The system records each event in the audit log with correct details, and logs are retained for at least 12 months.

TC-36 (Nathan)	FR-36	Verify that the health dashboard displays realtime server uptime, API response times, and active user sessions.	Log in as an administrator and open the health dashboard.	The dashboard shows live metrics for server uptime, API response times, and active user sessions.
TC-37 (Nathan)	FR-37	Verify that the health dashboard displays an alert when a monitored metric exceeds its defined threshold.	Simulate a monitored metric exceeding its confugured threshold while the health dashboard is open.	An alert appears on the dashboard identifying the metric that exceeded the threshold.
TC-38 (Nathan)	FR-38	Verify that the system logs login attempts, project updates, and application errors with a timestamp and user ID.	Perform a login, update a project, and trigger and application error on a test account.	Each event is recorded in the system log with a correct event type, timestamp, and associated user ID.
TC-39 (Nathan)	FR-39	Verify that administrators can search and filter log entries by date, user ID, and event type.	Log in as an administrator, open the log viewer, and apply filters for date range, user ID, and event type.	The log viewer returnss only entries matching all applied filter creteria.
TC-40 (Nathan)	FR-40	Verify that data from the health dashboard and the log viewer can be exported to external monitoring tools.	From the health dashboard and log viewer, initiate a data export and direct it to an external monitoring tool.	The export completes successfully and the data is received and readable in the external monitoring tool/

Non-Functional Requirements Test Cases

Test Case #	Related Requirement	Test Description	Test Steps	Expected Result
TC-01 (Alen)	NFR-1	Verify AI generates a complete project plan and product list within five seconds under reasonable load	1. Ensure system is under normal load 2. Open a stopwatch or timer or monitor network traffic to see response time/latency 4. Submit and monitor closely 5. Repeat variable number of times to check consistency	All variable attempts should return completed plans within five seconds from submission to generation. No single attempts shall exceed in-place thresholds
TC-02 (Alen)	NFR-2	Verify pricing, availability, and SKU properly map to live Home Depot site data without experiencing a maximum fifteen minute delay	1. Note current price of interested SKU 2. Wait for a price change to happen (or simulate one if not possible) 3. Now refresh prices in the materials and SKU page 4. Get the timestamp of change and align that with the timestamp from Home Depot's price change	Updated price appears on Materials and SKU page within 15 minutes of catalog change.
TC-03 (Alen)	NFR-3	Verify that system meets accessibility standards	1. Run website through an accessibility testing software 2. Try navigating site using just the keyboard	There should be zero WCAG 2.1 Accessibility violations. There should also be quick and easy access without

			3. Attach a screen-reader and ensure that all components of website have alt-text attached 4. All color contrasts should be appropriate and up to code	needing a mouse to navigate the site. Colors follow proper contrasts as needed.
TC-04 (Prajit)	NFR-4	Verify that user data is encrypted using AES-256.	Upload a project photo and attempt to access database storage using admin information.	File is unreadable and is displayed as encrypted code until decrypted by the necessary key.
TC-05 (Prajit)	NFR-5	Validate security rules for customer and admin logins.	Try to log into the Store Manager account and attempt a customer login using Google.	The Store Manager is prompted for an authentication code (MFA); customer is directed to the Google Authenticator portal.
TC-06 (Prajit)	NFR-6	Make sure the system is stable with 50,000 active users.	Use a tool and simulate 50,000 active sessions.	The system maintains a response time faster than 5 seconds.
TC-07 (Estrella)	NFR-7	Verify system uptime availability.	Monitor uptime during the testing period.	System maintains 99% availability.
TC-08 (Estrella)	NFR-8	Verify usability requirements.	User attempts to create a project plan.	System allows completion within three steps.
TC-09 (Estrella)	NFR-9	Verify session recovery after service interruption.	Simulate temporary service failure.	System restores user session without losing saved project.

TC-10 (Estrella)	NFR-10	Verify modular architecture maintainability.	Update or restart service module.	Systems continue functioning without affecting other components.
N-TC-11 (Nathan)	NFR-11	Verify that the system generates a health alert within 30 seconds of detecting a critical service failure.	Simulate failure of a critical component such as the AI engine or pricing integration and monitor the alerting system.	The system generates and displays a health alert withing 30 seconds of detecting the failure.
N-TC-12 (Nathan)	NFR-12	Verify that system logs and audit records are retained for at least 12 months.	Review stored logs and audit records in the logging system and check that records older than 12 months are still available.	System logs and audit records remain accessible for a minimum of 12 months.
N-TC-13 (Nathan)	NFR-13	Verify that sensitive user data is encrypted both in transit and at rest.	Upload a project photo, submit location data, and inspect transmitted and stored data using security testing tools and database access.	Sensitive user data is encrypted during transmission and is stored in the encrypted form in the system.
N-TC-14 (Nathan)	NFR-14	Verify that the system complies with data privacy regulations when collecting and processing user data.	Create a user account, provide consent preferences, submit personal data, and review whether the system requests consent, stores preferences, and allows changes	The system collects and processes user data only after user consent, records consent preferences, and allows the user to review or modify those preferences in

			to privacy settings.	compliance with privacy regulations.
TC-15 (Prajit)	NFR-15	Sync speed is verified between web and mobile interfaces.	Save a renovation plan on the web and quickly open the Home Depot mobile app on the phone.	The saved plan appears in the list of the user's projects within 2 seconds.

Requirements Traceability Matrix (RTM)

Functional Requirements RTM

Requirement	TC-01	TC-02	TC-03	TC-04	TC-05	TC-06	TC-07	TC-08	TC-09	TC-10
FR-01 (Alen)	X									
FR-02 (Alen)		X								
FR-03 (Alen)			X							
FR-04 (Alen)				X						
FR-05 (Alen)					X					
FR-06 (Alen)						X				
FR-07 (Alen)							X			
FR-08 (Alen)								X		
FR-09 (Alen)									X	
FR-10 (Alen)										X

Requirement	TC-11	TC-12	TC-13	TC-14	TC-15	TC-16	TC-17	TC-18	TC-19	TC-20
FR-11 (Prajit)	X									
FR-12 (Prajit)		X								

FR-13 (Prajit)			**X**							
FR-14 (Prajit)				**X**						
FR-15 (Prajit)					**X**					
FR-16 (Prajit)						**X**				
FR-17 (Prajit)							**X**			
FR-18 (Prajit)								**X**		
FR-19 (Prajit)									**X**	
FR-20 (Prajit)										**X**

Requirement	**TC-21**	**TC-22**	**TC-23**	**TC-24**	**TC-25**	**TC-26**	**TC-27**	**TC-28**	**TC-29**	**TC-30**
FR-21 (Estrella)	**X**									
FR-22 (Estrella)		**X**								
FR-23 (Estrella)			**X**							
FR-24 (Estrella)				**X**						
FR-25 (Estrella)					**X**					
FR-26 (Estrella)						**X**				
FR-27 (Estrella)							**X**			
FR-28 (Estrella)								**X**		
FR-29 (Estrella)									**X**	
FR-30 (Estrella)										**X**

Requirement	**TC-31**	**TC-32**	**TC-33**	**TC-34**	**TC-35**	**TC-36**	**TC-37**	**TC-38**	**TC-39**	**TC-40**

FR-31 (Nathan)	**X**									
FR- 32 (Nathan)		**X**								
FR-33 (Nathan)			**X**							
FR-34 (Nathan)				**X**						
FR-35 (Nathan)					**X**					
FR-36 (Nathan)						**X**				
FR-37 (Nathan)							**X**			
FR-38 (Nathan)								**X**		
FR-39 (Nathan)									**X**	
FR-40 (Nathan)										**X**

Non-Functional Requirements RTM

Requirement	**TC-01**	**TC-02**	**TC-03**
NFR-01 (Alen)	X		
NFR-02 (Alen)		X	
NFR-03 (Alen)			X

Requirement	**TC-04**	**TC-05**	**TC-06**	**TC-15**
NFR-04 (Prajit)	X			
NFR-05 (Prajit)		X		
NFR-06 (Prajit)			X	
NFR-15 (Prajit)				X

Requirement	TC-11	TC-12	TC-13	TC-14
NFR-11 (Nathan)	X			
NFR-12 (Nathan)		X		
NFR-13 (Nathan)			X	
NFR-14 (Nathan)				X

Requirement	TC-07	TC-08	TC-09	TC-10
NFR-07 (Estrella)	X			
NFR-08 (Estrella)		X		
NFR-09 (Estrella)			X	
NFR-10 (Estrella)				X

Test Execution on MVP and Results Documentation

TC#	TC Steps	Requirements Tested	Written By	Tested By	Pass / Fail
TC-1	1. Navigate to Create Account page 2. Enter a valid email and password 3. Click “Create Account” 4. Verify that the user is successfully logged in and redirected to the main application page	Supports foundational account functionality required for FR-21 (user data persistence and retrieval)	Nathan Tennyson	Alen	PASS

TC-2	1. Navigate to Login or Create Account page 2. Enter an invalid email (missing "@") 3. Enter any password 4. Attempt to submit form	Input validation and error handling for user account creation (supports system reliability)	Nathan Tennyson	Alen	PASS
TC-3	1. Log in with a valid account 2. Create a new project 3. Log out of the system 4. Log back in with the same account 5. Navigate to Projects page 6. Verify saved project is still present	FR-21	Nathan Tennyson	Alen	PASS
TC-4	1. Login with a valid account 2. Navigate to the plan page 3. Click any of the suggested project plans below 4. Verify that the system has extracted project scope and given the necessary material recommendations and steps according to the description	FR-11	Prajit	Nathan	PASS
TC-5	1. Login with valid account	FR-13	Prajit Alexander	Nathan	PASS

	2. Navigate to the plan page 3. Create a new project 4. Go to cost estimate 5. Add necessary building items to cart 6. View cart 7. Enter a valid zip code from Texas, California, or Florida 7. Validate that total costs reflect taxes for specific regions and environments according to the zip code (Florida: 7.03%, California: 8.83%, Texas: 8.25%)				
TC-6	1. Login with valid account 2. Create a new project 3. Go to cost estimate 4. Pick necessary building items 5. View cart 6. Toggle on the Pro-Loader assistance switch 7. Verify the availability of Pro-Loader slots for in-store pickup.	FR-14	Prajit Alexander	Nathan	PASS

TC-7	1. #TC-01 2. Now, within the product swap page, click on the AI modal and add in “replacing an electrical outlet” 3. Review that the steps are proper and comply with safety code and standards as appropriate 4. Make sure that steps also tie to the SKU in question	FR-09	Alen Jo	Nathan	PASS
TC-8	1. #TC-01 steps 2. Review the materials list generated from TC-01 3. Make sure that the displayed SKUs align with what Home Depot has	FR-02	Alen Jo	Nathan	PASS
TC-9	1. #TC-01 2. Go to the materials and SKU page 3. Add specifications such as room length, width, volume, etc.	FR-04	Alen Jo	Nathan	PASS

	4. Review recommended quantities based off room specs through some manual verification				
TC-10	Save project from results page.	FR-21	Estrella Avila	Estrella Avila	PASS
TC-11	Click project card to navigate.	FR-21	Estrella Avila	Estrella Avila	PASS
TC-12	Delete project from dashboard.	FR-21	Estrella Avila	Estrella Avila	PASS

Deployment Plan

Deployment Strategy

The BuildSmart system will follow a staged deployment approach aligned with its cloud-native architecture on Amazon Web Services (AWS). Deployment will occur after successful completion of system testing and User Acceptance Testing (UAT), ensuring all critical functional and non-functional requirements are validated before release.

The system will utilize a continuous integration and continuous deployment (CI/CD) pipeline integrated with GitHub to automate the build, test, and deployment process. Docker containers will be used to package the application and ensure consistency across environments.

The deployment process will follow a progressive rollout strategy, beginning with a staging environment and gradually transitioning to production. This minimizes risk and allows for early detection of issues before full public release.

Environment Setup

BuildSmart operates across three primary environments: development, testing (staging), and production. Each environment serves a distinct purpose in supporting the software lifecycle while ensuring consistency and reliability across deployments.

The development environment consists of local developer machines where team members build and test features using React Native for the frontend and Python with FastAPI for the backend. This environment is primarily used for initial feature development and unit testing. It is connected to development-level AWS resources, allowing developers to simulate integrations without affecting live systems.

The testing (staging) environment is hosted on AWS and is designed to closely mirror the production environment. This environment is used for integration testing, system testing, and User Acceptance Testing (UAT). It includes core system components such as the PostgreSQL database, Redis caching layer, and external API integrations. By replicating production conditions, the staging environment allows the team to validate real-world workflows and ensure system stability before release.

The production environment represents the live system deployed on AWS infrastructure. It utilizes a combination of services including containerized backend services (via Docker on AWS compute resources), Amazon S3 for file storage, PostgreSQL (RDS) for persistent data, and Redis for caching. Authentication is handled through AWS Cognito, while AWS CloudWatch is used for system monitoring and logging. This environment is configured for high availability, scalability, and secure access.

Release Process

The BuildSmart release process follows a structured pipeline to ensure code quality, system stability, and alignment with stakeholder expectations.

Development begins with code integration, where team members push updates to the GitHub repository. All changes are reviewed and merged into the main branch to maintain code integrity and traceability.

Once integrated, the system undergoes automated testing through a continuous integration pipeline. Unit tests (using Pytest and Jest), as well as integration and regression tests, are executed automatically. Any build that fails these tests is rejected, ensuring that only stable code progresses further.

After successful testing, the application moves to the build and containerization phase, where it is packaged into Docker containers. This ensures that all dependencies and configurations are consistent across environments.

The containerized application is then deployed to the staging environment, where system testing and User Acceptance Testing (UAT) are conducted. This step validates that the system meets both technical and business requirements.

Following successful validation, stakeholder approval is obtained from Home Depot representatives before proceeding to production.

During production deployment, the application is released to the live AWS environment. At this stage, monitoring tools such as AWS CloudWatch are activated to track system health, performance, and usage.

Finally, post-deployment monitoring ensures continued system reliability. Logs, performance metrics, and alerts are continuously observed, and any critical issues trigger immediate responses such as rollback procedures or hotfix deployments.

System Accessibility

The BuildSmart system is designed to support multiple access points, ensuring usability for both customers and internal stakeholders.

Users primarily interact with the system through web access, where the application is delivered via a browser-based interface. The frontend communicates securely with backend services through HTTPS, enabling users to generate project plans, view recommendations, and manage their projects.

In addition to web access, BuildSmart supports mobile access through React Native, allowing the platform to function across different mobile devices. This enables users to access their project plans, shopping lists, and recommendations on the go, providing flexibility and convenience.

The system also includes internal access for administrators and Home Depot associates. Authorized users can access tools such as the system health dashboard, log viewer, and project retrieval interfaces. These tools support operational monitoring, troubleshooting, and in-store customer assistance.

All user access is secured through AWS Cognito, which manages authentication, session control, and user identity. Cognito supports OAuth 2.0 and Multi-Factor Authentication (MFA), ensuring that the system meets required security and compliance standards.

Software Evolution

Future Enhancements

BuildSmart's current prototype establishes some of the core foundations of the system: project plan generation, SKU mapping, real-time catalog integration, cart building, and BI export. The path to full production will be phased across three stages following an initial release.

Phase 1 (0-3 months post release) will focus on hardening core features already done in the prototype. It may include improving accuracy of the Natural Language Processing pipeline, expanding more of the attach-rate recommendation system to take in more scenarios than just purchase data. There will also be stability fixes and improvements regarding performance latency as described in our non-functional requirements.

Phase 2 (3-6 months) will introduce more complex user-facing features. This can include a 3D AR project preview, photo and sketch upload with detection for layout, seasonal offers and materials, Pro Xtra member discount integration, and recurring order setup for consumables. The Find a Pro referral flow linking Bills of Materials to authorized service providers will also be completed here alongside split payment support across gift cards, credit cards, and financing plans.

Phase 3 (6-12 months) will finalize BuildSmart into a suitable production candidate. This phase covers the in-store associate portal, trending project recommendations, promotional offer display, the full Privacy Center with CCPA-compliant deletion workflows, and complete health dashboard and log viewer functionality for Home Depot IT. By the end of Phase 3, all functional and non-functional requirements outlined in this SOW will be fully satisfied.

Scalability Considerations

BuildSmart is designed with scalability as a concern to meet the demands of potentially 50,000 concurrent users during peak sales events as mentioned in NFR-6. The backend is containerized to support horizontal scaling, meaning additional containers can be spun up in response to traffic spikes without needing to manually scale out or intervene in the off chance that service disruptions occur due to too much traffic.

The catalog sync pipeline is designed to handle the growing SKU volume from Home Depot without degrading the 15-minute time-to-live window required by NFR-2. As the catalog grows, the pipeline batching and aggregation logic can tune independently from the rest of the system. This comes from the choice of making a modular architecture as set from NFR-10. The BI export pipeline will operate on its own 24-hour batch cycle, keeping demand exports isolated from user experience of BuildSmart.

Database layer will adopt read replicas and caching strategies as user volume increases to reduce load on primary data stores, particularly for tracking product adds or project creation. The system's 99.5% uptime target outlined in NFR-7 will be supported by load balancing across availability zones and automated failover mechanisms to ensure no single point of failure.

Technical Debt

As with any prototype, BuildSmart's current implementation carries known trade-offs that will need to be addressed before full production deployment. The most significant areas of technical debt are as follows.

Several features in our current prototype fail to be met due to restrictions from the Home Depot API. For example, being able to export the cart is not feasible at this time due to the immense security added by Home Depot and dynamic server-side rendering needed to generate a cart. This isn't possible with our current architecture and will need to be reworked to incorporate that. There's potentially mocked or partially integrated data sources for features like Delivery slot scheduling, Pro-Loader availability, and Pro Xtra member-specific pricing, which requires full API integrations to be made. The 3D AR model generation and photo-based layout detection are similarly stubbed out and will require substantial development investment to properly implement.

The current security posture meets minimum requirements but is not yet fully hardened for production. For example, regarding NFR-5, OAuth 2.0 is enforced for most of the logins through Firebase Authentication, but multi-factor has not been done due to costs associated and engineering work needed to be made for that. AES-256 encryption for photo uploads and personal data, as described in NFR-4 and NFR-13, awaits building the photo upload and personal data features before considering encryption. They'll also need to be audited for compliance across all storage layers.

The NLP project pipeline is functional but will need advanced fine-tuning and testing as it encounters more complex real-world scenarios and a wider variety of user input beyond the minimal testing of it. The attach-rate recommendation engine also needs more variables to work with apart from just purchasing data to pick up additional nuances in what products should be recommended and not just commonly bought complements.

In addition, certain features promised could not be delivered due to the time constraints given in the project or not well-defined requirements. For example, the feature to filter by brand, price, and availability is not a concrete requirement and would need further discussion or explanation to feasibly to create the feature.

Finally, current test coverage, based off the test cases, reflects the minimum viable product as done by some of the requirements. Therefore, before Phase 1 can begin, TAJA co will need to expand its test cases to cover scenarios, such as catalog sync, BI export, and cart generation pipelines.

Maintenance Strategy

TAJA Co will maintain BuildSmart through a structured release cadence following semantic versioning. Patch releases will address bug fixes and minor corrections, minor releases will introduce new features and non-breaking changes, and major releases will reflect significant architectural changes or breaking updates that requires a coordinated effort by Home Depot.

Scheduled maintenance windows will be communicated to Home Depot in advance and timed to minimize disruption to users. Emergency patches addressing critical service failures or security vulnerabilities will be fast-tracked outside the normal release cycle and deployed as quickly as possible to maintain BuildSmart's uptime commitment and any SLAs (Service Level Agreements) established.

All changes to BuildSmart must pass existing test cases as well as any new test cases required by the change before deployment. In anticipation of this, TAJA Co will create a staging environment to mirror production for validating updates before they go live.

User feedback will be incorporated through a structured threshold. Features requested by many users will evaluate feasibility and inclusion into the current system for a potential future release candidate. A/B testing will be done on significant UX or feature changes to validate user receptiveness and usability before full rollout. Telemetry and the health dashboard outlined in FR-36 will provide ongoing visibility into system performance and user behavior. These measures will allow TAJA Co to prioritize future improvements and identify areas of the system that may need modifications or removal.

Glossary of Terms

Term	**Definition**
AI (Artificial Intelligence)	Technology that enables the system to simulate human decision-making and generate project recommendations.
API (Application Programming Interface)	A set of rules that allow different software systems to communicate and exchange data.
AR (Augmented Reality)	A technology that overlays digital project models onto real-word space through a mobile device camera.
Bill of Materials (BOM)	A complete list of materials, tools, and quantities needed for a project.
Consent Management	The process of collecting, storing, and updating user permission choices for data use.
Real-Time Inventory	Product availability that reflects current stock levels at specific store locations.
SKU	A unique stock keeping unit identifier assigned to a specific Home Depot product.
Natural Language Processing (NLP)	An AI technique that allows the system to understand and process user input written in everyday language.
PII (Personally Identifiable Information)	Any information that can identify a specific person, such as name, email, or location data.
Recommendation Engine	A system component that suggests related materials, tools, or alternative products based on the user's project.
Payment Processing	The system component responsible for securely handling transactions using credit cards, gift cards, or financing options.
Service Monitoring	The process of continuously tracking system components to detect failures, performance issues, or outages.
Product Catalog	A centralized database containing product details such as SKU, pricing, specifications, and availability.
Inventory Management System	A system used to track product stock levels, availability, and movement across stores and warehouses.
Cloud Architecture	A system design approach where computing resources, storage, and services are hosted on cloud infrastructure to allow scalability, reliability, and high availability.